BME555a Team Cardinal Project Report

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Team Cardinal

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1. Design for Manufacturability (DFM)

This section documents the design-for-manufacturability of the NeuroFoot System, a wearable plantar pressure-monitoring device for patients with diabetic peripheral neuropathy. The documentation follows ISO 13485:2016, Sections 7.3 (Design and Development) and 7.5 (Production and Service Provision), and is structured so that a new team member can review this material and continue the project without relying on institutional knowledge from the original developers. The section is divided into two subsections: design inputs, which trace stakeholder needs through to measurable product specifications, and design outputs, which document the technical artifacts, acceptance criteria, and manufacturing considerations required to move from prototype to production.

1.1. Design Input & Manufacturability Requirements

Design inputs define what the device must do and how well it must do it. Per ISO 13485 Section 7.3.3, design inputs must be derived from identified user needs, translated into verifiable product specifications, and reconciled with applicable regulatory and clinical requirements. This subsection documents the stakeholder analysis, the translation of unmet needs into features and specifications, the selected materials and tolerances, and the regulatory classification and predicate device research that govern the device's pathway to market.

1.1.a. Stakeholder Needs and Design Specifications

The NeuroFoot System is indicated for use in adult patients (18 years and older) with diabetic peripheral neuropathy who are at elevated risk for plantar ulceration due to reduced or absent protective sensation (IWGDF Risk Category 2–3). The device continuously monitors plantar pressure distribution across 16 independently sensed zones, ambient temperature and humidity within the footwear, and gait-related loading patterns during daily ambulation. When sensor readings exceed predefined thresholds, the system generates a real-time alert notifying the user and caregiver. The device is indicated for Prescription Use only, under the supervision of a licensed physician or clinician.

The NeuroFoot System is intended to function as a non-invasive wearable monitoring and threshold-alerting system. The device does not diagnose, treat, or cure any medical condition, including diabetic foot ulcers. The system detects when a measured parameter has exceeded a predefined threshold and issues a warning signal. Clinical interpretation of that warning—including any diagnostic or treatment decision—is made solely by the supervising clinician. The physician, not the device, makes the diagnosis. The device is intended as an adjunct to standard diabetic foot care under clinician supervision.

Table 1. Stakeholder Register

Stakeholder	Clinical Role	Priority Rationale
Neuropathic Patient (IWGDF Risk 2-3)	Primary alert recipient; performs offloading in response to alerts; responsible for daily wear and charging	Direct bearing of harm from failure; highest priority per ISO 14971
Caregiver	Secondary alert recipient; assists with offloading when the patient cannot self-manage	ADA Standards of Care identify caregivers as essential in high-risk DFU populations
Clinician (Podiatrist, Endocrinologist)	Reviews longitudinal data during clinical encounters; modifies preventive care strategy	IWGDF 2023 recommends periodic pressure monitoring for high-risk foot surveillance
FDA CDRH	Regulatory reviewer for premarket submission	Defines regulatory constraints (21 CFR 890.5575, Class II)

The core design input traceability for the NeuroFoot system links each stakeholder's unmet need to one or more product features and measurable specifications. The traceability follows the structure required by ISO 13485:2016 Section 7.3, and is organized below by subsystem.

The following tables constitute the formal design input record. Each row traces a confirmed stakeholder unmet need through to one or more measurable product specifications. The tables are organized by subsystem, and each is followed by a description of the listed design features.

Table 2. Subsystem A: Plantar Pressure Sensing

Stakeholder & Unmet Need	Design Feature	Product Specification	Rationale
Patients with LOPS, “Loss of Protective Sensation,” are unable to perceive sustained plantar loading, allowing tissue damage to develop without awareness.	Multi-zone pressure sensing integrated into the insole	The System incorporates a pressure-sensing array with at least 16 sensing locations distributed across key plantar regions, including the toes, metatarsal heads, midfoot, and heel.	Clinical guidelines identify elevated plantar pressure, particularly at the metatarsal heads and forefoot, as a primary risk factor for tissue injury. Multi-zone sensing enables detection of localized high-pressure regions that would otherwise go unnoticed by the patient.
Patients require objective information to understand when pressure levels may be harmful.	Quantitative pressure measurement	The System measures plantar pressure across all sensing locations with sufficient resolution to distinguish clinically meaningful changes in loading during standing and walking.	Quantitative measurement enables the system to detect changes in loading patterns that are imperceptible to patients with neuropathy, supporting timely awareness and response.

Patients need timely feedback to interrupt potentially harmful loading.	Real-time pressure monitoring and alerting	The System continuously evaluates pressure data and provides alerts when sustained loading exceeds predefined thresholds associated with increased mechanical stress.	Timely feedback supports behavioral offloading, which is a key component of preventive care strategies for patients at risk of foot complications.
Sensor performance must remain reliable over time to avoid missed or inaccurate alerts.	Calibration and performance stability controls	The System includes automatic calibration routines and is designed to maintain consistent pressure measurement performance over repeated use.	Maintaining measurement accuracy is essential to ensure that clinically relevant loading is not underestimated and to prevent missed alerts.
Clinicians require information on where pressure is concentrated over time.	Data logging and visualization	The System records pressure data across all sensing locations and makes it available for review over time.	Longitudinal pressure data support clinical evaluation and align with recommendations for ongoing monitoring of high-risk patients.

The pressure-sensing subsystem addresses the primary clinical need identified in the stakeholder analysis. Patients with loss of protective sensation cannot perceive sustained plantar loading, so the system provides what their nervous system cannot: continuous, quantitative measurement of the force applied to each region of the foot. Each of the five design features in this subsystem addresses a different aspect of that function. Multi-zone sensing refers to the physical hardware. The 16-zone FSR array is distributed across the toes, metatarsal heads, midfoot, and heel. Quantitative pressure measurement refers to the measurement resolution of that hardware, ensuring the ADC output can distinguish between safe and dangerous loading levels. These two features are related but distinct: one defines where the system measures, and the other defines how precisely.

Real-time monitoring and alerting is the software layer that continuously evaluates the incoming pressure data and triggers WARN or CRITICAL alerts when thresholds are exceeded. This is the feature that delivers the threshold-based warning function during daily use; the supervising clinician interprets the alert and determines any appropriate clinical response. Calibration and performance stability refer to the routines that keep the pressure readings accurate over time, so that sensor drift or aging does not gradually erode the alert thresholds. Data logging and visualization serve a different audience. It records pressure trends over days and weeks so that clinicians can review the data during follow-up visits, rather than relying solely on the patient's real-time alerts.

Table 3. Subsystem B: Environmental Sensing

Stakeholder & Unmet Need	Design Feature	Product Specification	Rationale

Patients with LOPS, “Loss of Protective Sensation,” are unable to perceive prolonged moisture or unfavorable thermal conditions within the shoe, which may contribute to skin breakdown.	Environmental sensing within footwear	The System monitors temperature and humidity within the shoe environment during use.	Environmental conditions, such as elevated humidity and temperature, are associated with skin maceration and an increased risk of tissue damage. Providing this information helps users be aware of potentially unfavorable conditions.
Clinicians and users may expect temperature data to reflect physiological indicators of inflammation.	Defined scope and labeling of environmental data	Environmental measurements are clearly presented as ambient conditions within the shoe and are not intended to represent plantar skin temperature.	Clinical guidelines recommend skin temperature monitoring as an early indicator of inflammation; however, the System measures only ambient conditions. Clear labeling prevents misinterpretation and ensures appropriate use of the data.

The environmental sensing subsystem has two design features, and the second exists specifically to prevent misuse of the first. Environmental sensing within footwear refers to the HS300x temperature and humidity sensor mounted on the Arduino board between the sandal layers. It monitors ambient conditions inside the sandal during use. Elevated humidity and temperature are associated with skin maceration, so this data adds context to the pressure readings. Defined scope and labeling address the risk that clinicians or patients might interpret these readings as skin temperature measurements, which clinical guidelines use as an indicator of inflammation. Because the HS300x measures ambient air, not the plantar surface, the companion app labels the readings as ambient to prevent misinterpretation.

Table 4. Subsystem C: Wireless Communication and Alerting

Stakeholder & Unmet Need	Design Feature	Product Specification	Rationale
Patients require timely, perceptible alerts to respond to harmful loading; caregivers may need to assist when patients cannot respond independently.	Wireless alert delivery and user notification	The System transmits data to a paired smartphone and provides real-time alerts through visual, auditory, or haptic feedback.	Timely alerts enable users to take corrective action, such as offloading pressure, which is a key component of preventive foot care.
The system must remain reliable and avoid silent failure, particularly in patients	Battery monitoring and status alerts	The System provides clear indications of battery status and alerts the user when the battery is low.	Preventing unexpected loss of function is critical to maintaining continuous monitoring and ensuring user safety.

unable to perceive physiological warning signals.			
Wireless data transmission must be secure to prevent unauthorized access.	Secure data communication	The System incorporates standard wireless security measures appropriate for the transmission of medical data.	Ensuring data security supports regulatory requirements and protects patient information when data is transmitted or stored.

The wireless subsystem has three design features that serve different purposes. Wireless alert delivery is the core function: it transmits sensor data over BLE to a paired smartphone and delivers alerts via visual, auditory, and haptic feedback so patients and caregivers can respond to dangerous loading in real time. Battery monitoring and status alerts exist to prevent the scenario in which the device silently dies mid-use, and the patient walks unmonitored for the rest of the day. Because this population cannot feel when something is wrong, a dead device is functionally invisible to them, which makes low-battery warnings a safety-critical feature rather than a convenience. Secure data communication covers the BLE encryption and pairing protocols that protect transmitted medical data, which is a regulatory expectation for any wireless Class II device.

Across all three subsystems, the ten design features are distinct. The closest overlap is between multi-zone pressure sensing and quantitative pressure measurement in Subsystem A, but these address different aspects of the same sensor system, spatial coverage versus measurement resolution, and both are necessary to satisfy the clinical requirement of detecting localized dangerous loading with adequate precision.

Table 5. Subsystem D: Gait and Activity Monitoring

Stakeholder & Unmet Need	Design Feature	Product Specification	Rationale
Abnormal gait patterns may contribute to increased cumulative plantar stress, which cannot be fully captured by pressure measurements alone.	Motion sensing for gait context	The System includes motion-sensing capabilities to capture gait-related parameters during ambulation.	Gait information provides context for interpreting pressure data, helping to distinguish between different loading conditions during movement.
Clinicians require interpretable data to understand patient loading patterns over time.	Integration of gait and pressure data	Gait-related information is recorded alongside pressure data and made available for review.	Combined data support a more complete understanding of patient movement and loading behavior, aiding clinical evaluation.

The Gait and Activity Monitoring subsystem has two design features that work together to provide movement context for the pressure data collected by the system. Motion sensing for gait context captures gait-related parameters during ambulation so the system can distinguish between different loading conditions, such as separating a normal heel strike from an abnormal compensatory pattern that concentrates stress on a vulnerable plantar region. Without this context, a pressure reading alone cannot reveal whether a spike was incidental or the result of a recurring loading pattern that will repeat with every step, a distinction the supervising clinician can evaluate using the logged data. Integration of gait and pressure data ensures that motion information is recorded alongside pressure measurements and made available for clinical review, giving clinicians longitudinal movement and loading data over time, rather than isolated snapshots of peak pressure events. The clinician uses this data to inform their assessment; the device does not interpret or diagnose the significance of the recorded patterns.

Table 6. Subsystem E: System-Level and Safety Specifications

Stakeholder & Unmet Need	Design Feature	Product Specification	Rationale
The device must operate safely within the physical and environmental conditions of daily use.	Robust system design for wearable use	The System is designed to operate reliably within a footwear environment, including exposure to movement, temperature variation, and moisture.	Safe and reliable operation in real-world conditions is essential for continuous use and user safety.
Software functionality must be reliable and appropriate for its clinical role.	Software-based monitoring and alerting	The System software processes sensor data and generates alerts based on predefined conditions.	The software supports monitoring and alerting functions and does not perform autonomous diagnosis or treatment decisions.
The device must be usable by patients without specialized training.	Usability-focused design	The System is designed with clear and interpretable alerts and user interactions that can be understood without extensive training.	Usability is critical for ensuring that users correctly interpret alerts and take appropriate action, particularly in populations with potential physical or cognitive limitations.

The System-Level and Safety Specifications subsystem has three design features that address system-level reliability, software boundaries, and usability. Robust system design for wearable use ensures the device operates reliably within a footwear environment, accounting for the movement, temperature variation, and moisture exposure inherent to daily ambulation. These are baseline requirements for any wearable medical device, but they are particularly important here because a device that fails intermittently in real-world conditions offers no meaningful safety benefit to a population that cannot perceive when monitoring has lapsed. Software-based monitoring and alerting defines the functional scope of the system software, which processes sensor data and generates alerts based on predefined thresholds but does not perform autonomous diagnosis or treatment decisions. This boundary is a regulatory requirement: the device issues warnings when thresholds are exceeded, and the supervising clinician is solely responsible for any resulting diagnosis or treatment decision. Usability-focused design ensures that alerts and user interactions are interpretable without specialized training, which is essential given that the target population may have physical or cognitive limitations that affect their ability to respond to complex or ambiguous feedback.

1.1.b. Materials, Tolerances, and Production Constraints

The material selections for the NeuroFoot system are driven by two competing requirements: the device must be mechanically robust enough to survive repeated compressive loading inside footwear, and it must remain thin and compliant enough that the patient perceives it as a normal sandal rather than a medical instrument. The printed circuit board uses a rigid FR4 substrate, which is viable in this application because the surrounding EVA foam absorbs the bending stresses that would otherwise crack a rigid board during walking. The sandal body itself is constructed from two layers of thick EVA foam that serve simultaneously as the mechanical enclosure, the load-distribution layer above the sensors, and the protective housing for the embedded electronics. Foam durometers must be documented and held consistent across units, because softer foam dampens peak pressures and can cause sensors to underread high-risk zones. All materials in direct contact with the plantar surface, including the foam top layer and any resealing adhesive, must be biocompatible and hypoallergenic. Diabetic skin is more fragile and prone to breakdown than healthy tissue, and chemical irritation at the foot surface is itself a direct ulcer risk.

The pressure-sensing array consists of 16 individual MakerHawk RP-C thin film force-sensitive resistors (RP-C type, 20 g to 2 kg range) that the team soldered onto a custom circuit board and positioned at anatomically relevant planar zones. This is not a pre-made insole sensor; each FSR was individually placed and wired to achieve the 16-zone layout described in Section 1.1.a. The sensors are rated for a force range of approximately 0 to 100 N per site (peak approximately 700 kPa at the heel) and must survive millions of compression cycles over the device lifetime. Each FSR zone is paired with a 10 kΩ pull-down resistor in a voltage divider configuration, which produces a usable ADC range of approximately 930 counts across the clinically relevant force range. The team initially used 100 kΩ resistors, which compressed the useful ADC range to roughly 270 counts; the switch to 10 kΩ provided significantly better resolution for distinguishing between normal, elevated, and dangerous pressure levels. Piezoelectric sensors were evaluated as an alternative but rejected because they respond only to dynamic pressure changes, not sustained static loading, which is the primary clinical concern in diabetic neuropathy.

The microcontroller is an Arduino Nano 33 BLE Sense Rev2, selected because it integrates an ATmega-class processor, onboard BLE 5.0 radio, a BMI270 6-axis IMU (accelerometer and gyroscope), and an HS300x temperature and humidity sensor on a single 45 × 18 mm board. This integration eliminates the need for separate breakout modules for wireless communication, motion sensing, and environmental monitoring, which keeps the overall footprint small enough to embed inside the sandal midsole. The CD74HC4067 16-channel analog multiplexer routes all 16 FSR zones through a single analog input pin on the Arduino, since the Nano 33 BLE Rev2 has only 8 analog inputs. All 16 channels are utilized, leaving no unused pins. Voltage compatibility with the Arduino's 3.3 V logic must be confirmed, and the multiplexer's channel switching time of approximately 100 nanoseconds is well within the 100-microsecond settle time used in firmware. Power is planned to be supplied by a small LiPo battery positioned in the midfoot/arch region, away from high-load zones, and charged via USB-C or a magnetic connector that is clearly labeled for users with reduced dexterity.

All bulky rigid components, including the Arduino, multiplexer, battery connector, and FPC breakout board, are positioned in the midfoot and arch region of the sandal, which is the lowest load-bearing anatomical zone. This placement prevents hard spots from forming under the heel or metatarsal heads, where ulcer risk is highest. The electronics are sandwiched between the two structural foam layers, and the cavity dimensions are sized to accommodate the Arduino board, breakout board, and wiring within the midsole.

The system's performance criteria are organized around three functions: pressure sensing, gait-phase detection, and alerting. The sampling rate is set at 10 to 50 Hz, which is sufficient for gait analysis; the Arduino and CD74HC4067 combination must sustain this rate continuously during operation. The 16 FSR sensors are mapped to anatomically relevant zones: heel medial, heel central, heel lateral, medial arch, lateral midfoot, 1st through 5th metatarsal heads, hallux, 2nd/3rd toe cluster, 4th/5th toe cluster, central forefoot, lateral forefoot border, and lateral midfoot border. The 16-zone layout was selected on the basis of a risk analysis. The five highest-ulcer-risk anatomical sites (1st and 5th metatarsal heads, hallux, heel, and lateral midfoot) each require independent sensing to avoid masking focal overload through zone averaging. Fewer zones (for example, 8) would merge adjacent high-risk sites into single readings, reducing the system's ability to detect localized threshold exceedances and increasing the risk of a missed alert. The predicate device (SmartStep, K060150) uses single-threshold detection; the 16-zone array improves spatial resolution without introducing new risks, consistent with the substantial equivalence rationale presented in Section 1.1.c.

The measurement goal is to track the duration of pressure loading per zone, not instantaneous peak force, because the clinical risk factor for diabetic foot ulcers is sustained pressure over time rather than momentary spikes. Pressure accumulation is gated to stance phase only, using IMU-derived gait phase detection. The BMI270 onboard accelerometer and gyroscope provide gait phase detection (heel strike, toe-off, mid-stance), fall detection, and activity context (standing versus walking versus sitting), which the firmware uses to adjust alert logic accordingly. An alert is triggered when a single zone exceeds a continuous loading duration threshold, for example 30 or more minutes during stance phase, and that threshold must be clinically referenced against the published diabetic foot ulcer pressure-time literature. The alert output is an onboard buzzer and/or vibration motor that operates independently of BLE, so that alerts function without a connected phone. BLE transmission to the companion app provides secondary caregiver notification. Visual-only or app-only alerts are insufficient for this patient population, because many patients with neuropathy also have visual impairments or may not carry their phone at all times. The battery must provide a minimum of 8 to 12 hours of continuous operation per charge to cover a full day of wear.

One important measurement limitation must be acknowledged in all clinical documentation: the FSR readings represent relative pressure distribution, not calibrated absolute plantar pressure. The foam compliance layer that sits above the sensors introduces a measurement offset, and the voltage-divider circuit produces a nonlinear response. The system is designed to detect relative changes and sustained loading patterns rather than to serve as a calibrated pressure-measurement instrument.

The critical tolerances are derived from the requirement that sensor position must correspond to the intended anatomical zone, that the foam layer must transmit pressure uniformly, and that the electronics must remain mechanically stable inside the sandal over the product lifetime. Sensor placement must be held to within ±2 to 3 mm, because misalignment shifts which anatomical zone is being measured and invalidates zone-specific ulcer risk warnings. PCB fixation inside the foam must be rigid and repeatable, with no migration during walking. Board migration shifts sensor positions away from intended anatomy with no visible indicator to the patient or caregiver, so epoxy bonding or a snug foam cavity with adhesive retention is required. Foam layer thickness above the sensors must be uniform to within ±0.3 mm across the full insole surface, because uneven thickness creates artificial pressure hotspots unrelated to actual plantar load and invalidates the foam compliance correction factor applied during calibration. All solder joints must withstand repeated compressive and shear loading over the product lifetime, with strain relief required where component leads exit bodies. Rigid components must not create hard focal points perceptible through the foam. Finally, all bulky rigid components (Arduino, multiplexer, battery connector) must remain in the midfoot/arch region to prevent hard spots under the heel or metatarsal heads where ulcer risk is highest.

The production constraints reflect the fact that the NeuroFoot prototype is assembled by hand, and that several steps in the assembly process are not easily repeatable without standardization. Cutting and reassembling the foam sandal around the PCB requires documented cavity dimensions, PCB seating depth, alignment references, and a controlled foam resealing method to ensure consistent sensor positioning across all units. Foot geometry varies significantly across the diabetic patient population, so production must support small, medium, and large variants of both the PCB layout and the sandal body. Alternatively, a design that accommodates trimming without severing PCB traces must be validated. The rigid FR4 PCB is a straightforward and low-cost choice relative to flexible PCB alternatives, but component height must be minimized to keep total sandal thickness within a footwear-compatible range without compromising the foam cushioning above the sensors.

Soldering introduces additional variability. SMD components benefit from stencil application and reflow oven processing, but hand soldering is feasible at prototype quantities and introduces joint quality variability that must be controlled through mandatory visual inspection and continuity testing before the board is embedded in foam. Foam resealing is another critical process: the adhesive type, quantity, and application method must be specified and controlled, because resealing consistency directly affects foam layer uniformity above the sensors and the long-term structural integrity of the assembly. All electronics must be conformal coated before embedding, because post-assembly sealing is not feasible once the PCB is enclosed within the sandal body. Foam absorbs sweat over time, and unprotected electronics will fail from moisture ingress.

The NeuroFoot bill of materials sources components from vendors whose quality management systems vary in scope and certification level. The MakerHawk RP-C thin film pressure sensors are manufactured by Shenzhen Ligan Technology Co., Ltd., a supplier that holds ISO 9001 general quality management certification but does not hold ISO 13485 medical device quality management certification. The same gap applies to most of the other components on the BOM, including the CD74HC4067 multiplexer and the LiPo battery cells, which are manufactured under general commercial quality systems rather than medical device quality systems. The Arduino Nano 33 BLE Sense Rev2 is produced by Arduino S.r.l. under ISO 9001, and while the onboard BMI270 and HS300x sensors are manufactured by Bosch Sensortec and Renesas respectively (both ISO 9001 certified), neither component is qualified under a medical device quality system. This certification gap means that incoming inspection at Stage 1 of the assembly process carries additional importance, because the team cannot rely on vendor-supplied certificates of conformance that meet ISO 13485 traceability requirements. Each lot of FSR sensors must be individually tested before use, and any component that does not pass the pre-solder or incoming inspection criteria defined in Section 1.2.d must be rejected regardless of the vendor's own quality claims. If the NeuroFoot system advances beyond the prototype stage toward commercial production, a formal supplier qualification process should be established in accordance with ISO 13485 Section 7.4, including documented supplier audits, approved supplier lists, and purchasing specifications that define acceptance criteria independent of vendor certification status.

Finally, fault detection must be implemented in firmware before deployment. A watchdog timer and heartbeat signal ensure that the system generates a distinct failure alert, separate from a pressure-overload alert, if sensors stop reading, firmware crashes, or the battery critically depletes. Silent device failure is a patient safety risk, not merely a quality issue, given that neuropathic patients have no sensory fallback to detect that monitoring has lapsed. The firmware implements a single-sensor dropout policy: if one or more individual zones return out-of-range readings (below noise floor or saturated) during operation, the affected channel is flagged as faulted and excluded from threshold comparison, and the fault is logged with a fault code. If the number of faulted channels exceeds three, meaning more than one high-risk anatomical region is unmonitored, the device escalates to a distinct sensor-fault alert, notifying the user and caregiver that monitoring coverage is degraded and the device requires inspection. Silently zeroing or ignoring a faulted channel is not permitted, as it could suppress an alert that would otherwise have triggered on that zone.

1.1.c. Regulatory Classification and Predicate Device

The most relevant predicate device for the proposed smart insole system is the SmartStep System, developed by Andante Medical Devices Ltd. (Yoqneam Illit, Israel). The SmartStep System received FDA 510(k) clearance on February 23, 2006, under premarket notification number K060150, following a traditional 510(k) review with K023161 (an earlier SmartStep model) as its own predicate (U.S. Food and Drug Administration, “K060150”). The device is regulated as a Class II medical device under 21 CFR 890.5575 and is assigned to the Physical Medicine review panel under product code IRN, classified as a Powered External Limb Overload Warning Device (U.S. Food and Drug Administration, “21 CFR 890.5575”). The official FDA clearance documentation is publicly accessible through the FDA 510(k) Premarket Notification Database.

The SmartStep System consists of three components: a flexible pressure-sensing insole worn inside the patient’s shoe, a belt-worn Control Unit that processes sensor data and generates alerts, and PC-based Software for clinical review and threshold configuration. The system is designed to sense the weight applied to the plantar surface of the foot during rehabilitation and to alert the user and/or therapist with an alarm when the weight exceeds a pre-selected value. It is indicated for Prescription Use under the supervision of a physician or physiotherapist. The device complies with IEC 60601-1, IEC 60601-1-2, and AAMI/ISO 14971-1 (U.S. Food and Drug Administration, “K060150”).

This predicate was selected over the previously considered Qualisys Clinical System (K171547, 21 CFR 890.5360, product code LXJ) because the SmartStep shares the same fundamental device purpose, sensing modality, form factor, and regulatory classification as the proposed device. The Qualisys system is an optical motion capture system intended for laboratory-based gait analysis. It does not monitor plantar pressure, does not use a wearable insole, and does not generate patient-facing overload warnings. By contrast, the SmartStep is a wearable pressure-sensing insole that alerts patients to dangerous plantar loading, a directly analogous function to the proposed device. The IRN product code accurately captures the core clinical function of both devices.

The Indications for Use of the SmartStep System state that it is intended to sense the amount of weight applied to the plantar surface of the foot during rehabilitation and to alert the user and/or therapist with an alarm when the weight exceeds a pre-selected value. The device is indicated for use in adults undergoing rehabilitation who require monitoring of partial weight-bearing on the lower extremity. It is for Prescription Use under the supervision of a physician or physiotherapist (U.S. Food and Drug Administration, "K060150"). These indications define the patient population and the clinical context in which the system operates. The Intended Use describes the device's fundamental purpose: to detect dangerous mechanical loading on the plantar foot and provide a warning that prompts corrective action, thereby preventing injury during the rehabilitation process.

The proposed device has distinct but related Indications for Use. It is intended for adults with diabetic peripheral neuropathy who are at elevated risk for plantar ulceration due to reduced protective sensation. The device monitors plantar pressure distribution across 16 independently sensed zones (0–400 kPa, 50 Hz) and gait loading patterns (6-axis IMU, 50 Hz) during daily ambulation. An onboard environmental sensor (HS300x) also logs ambient temperature and humidity. When sensor readings exceed predefined thresholds, the system generates graded real-time alerts (OK / WARN / CRITICAL) notifying the user and caregiver. The device does not diagnose, treat, or cure any medical condition. Clinical interpretation of alerts and any resulting treatment decisions are made solely by the supervising clinician. The Intended Use is to function as a non-invasive wearable threshold-alerting system for prescription use under clinician supervision. The device is for Prescription Use under a clinician's supervision.

The SmartStep System serves as an appropriate Class II predicate because both systems share the central clinical function of detecting dangerous plantar loading and warning the user to take corrective action. Although the SmartStep monitors pressure during supervised rehabilitation and the proposed device monitors pressure during continuous daily ambulation, both systems are designed to generate real-time alerts when plantar stress exceeds clinically significant thresholds. The technological differences between the two devices do not alter the core intended use or introduce new safety concerns relative to the cleared predicate.

The primary technological differences are as follows. First, the proposed device uses a 16-zone pressure mapping array rather than single-threshold detection, providing higher spatial resolution for identifying focal overload zones. The underlying sensing technology remains resistive force sensing, and the higher resolution improves alert specificity without introducing new risks. Second, the proposed device includes an IMU-based gait analysis subsystem (BMI270, accel ±4g, gyro ±2000 dps) that provides contextual data including step count, cadence, and stance timing.

This motion data provides additional context for threshold gating by distinguishing static sustained loading from dynamic gait events. Third, integrating pressure and gait data into a graded composite risk score (OK / WARN / CRITICAL) provides clinical decision support consistent with SmartStep's threshold-based alerting. The risk score does not constitute an autonomous diagnosis; it prompts the user to take behavioral action, identical in function to SmartStep's audible alarm.by distinguishing stance phase (weight-bearing) from swing phase (non-weight-bearing), preventing false threshold accumulation.

Third, the device generates graded alerts (OK / WARN / CRITICAL) when thresholds are exceeded, identical in function to SmartStep's audible alarm. The alert does not constitute a diagnosis; it is a threshold notification that the supervising clinician interprets. Fourth, the device includes an onboard environmental sensor (HS300x) that logs ambient temperature and humidity. These readings provide general environmental context and are recorded alongside pressure and gait data for clinician review.

Fifth, replacing SmartStep's belt-worn Control Unit and PC Software with a BLE-connected smartphone application modernizes the data display pathway using a well-established wireless standard present in numerous cleared Class II medical devices. Sixth, the proposed device uses a sandal form factor with the microcontroller (Arduino Nano 33 BLE Sense Rev2) housed between two structural layers of the sandal, rather than a separate belt-worn unit. This integration reduces the number of external components while maintaining the same non-invasive, externally worn profile. Seventh, extending monitoring from rehabilitation sessions to continuous daily wear does not introduce new safety risks, as the device remains non-invasive, externally worn, and supervised under prescription use.

Eighth, calibration: the SmartStep System uses a user-adjustable weight threshold set via PC software prior to use, referenced against the patient's prescribed weight-bearing limit (U.S. Food and Drug Administration, "K060150"). The proposed device uses a zone-level duration threshold calibrated against published diabetic foot ulcer pressure-time literature (see Performance Criteria, Alert Threshold). Both approaches set a single reference threshold per session; the proposed device's threshold is duration-based rather than instantaneous-force-based, consistent with the sustained-loading etiology of neuropathic ulcers. FDA reviewers should note that the calibration methodology differs in parameter (duration vs. instantaneous force) but is equivalent in structure (clinician-referenced, pre-set threshold that triggers a user alert). See Table 5 for the full substantial equivalence comparison.

The SmartStep System's FDA 510(k) clearance summary (K060150) and the device labeling submitted to FDA describe the calibration workflow at a high level: the clinician sets a single weight threshold through the PC-based software application prior to each use session, and the system generates an audible alarm when the measured plantar force exceeds that threshold. The clearance documentation does not include the full operating manual, and the SmartStep operating manual is not publicly available through the FDA 510(k) Premarket Notification Database or through Andante Medical Devices Ltd.'s public records. The team was therefore unable to obtain detailed step-by-step calibration instructions, recalibration intervals, or sensor maintenance procedures from the predicate device's documentation. This limitation is noted because a complete predicate comparison under 21 CFR 807.87(f) would ideally reference the predicate's labeling, including its instructions for use, to confirm that the proposed device's calibration methodology does not introduce new risks relative to the cleared device. The comparison presented in Table 5 is based on the information available in the FDA clearance summary, which identifies the core calibration parameters (clinician-set threshold, single reference value per session, duration-based versus force-based measurement) but does not describe operational details such as recalibration frequency, sensor drift management, or end-of-life criteria. If the predicate operating manual becomes available through a Freedom of Information Act request or direct contact with the manufacturer, the substantial equivalence analysis should be updated to include a detailed calibration methodology comparison.

These differences represent enhancements to the same core clinical function and do not raise new safety or effectiveness questions relative to the cleared predicate (U.S. Food and Drug Administration, “K060150”), thereby supporting a defensible substantial equivalence rationale.

The device design specifications are referenced with guidelines from relevant clinical organizations. The International Working Group on the Diabetic Foot (IWGDF 2023) identifies elevated and sustained plantar pressure as a primary modifiable risk factor for ulceration in neuropathic patients, and recommends pressure-offloading interventions when abnormal loading is detected (Bus et al.). The device’s 16-zone pressure monitoring and sustained-load alerting directly support this guideline. The IWGDF further recommends structured monitoring programs for high-risk patients, consistent with the device’s continuous ambulatory monitoring approach. The American Diabetes Association Standards of Care (2024) recommends comprehensive foot examinations for patients with peripheral neuropathy and emphasizes the role of patient education, appropriate footwear, and early intervention for abnormal foot mechanics. The device’s real-time biofeedback and behavioral prompting align with the ADA’s emphasis on patient engagement in preventive foot care. The American Podiatric Medical Association endorses wearable pressure-offloading technologies and routine screening for patients at risk of diabetic foot complications. The sustained-pressure detection and offloading prompts are consistent with APMA recommendations for reducing mechanical stress in at-risk populations.

Table 7. Substantial Equivalence Comparison: SmartStep System (K060150) vs. NeuroFoot

Feature / Function	Predicate: SmartStep System (K060150)	NeuroFoot (Proposed Device)	Substantial Equivalence
Device Classification	Class II, 510(k) cleared (K060150; predicate K023161)	Class II, 510(k) pathway (proposed predicate: K060150)	Same class and regulatory pathway
Regulation / Product Code	21 CFR 890.5575; Product Code IRN (Powered External Limb Overload Warning Device)	21 CFR 890.5575; Product Code IRN (Powered External Limb Overload Warning Device)	Identical regulation and product code
Intended Use	Sense the amount of weight applied to the plantar surface of the foot during rehabilitation,	Continuously monitor plantar pressure distribution and gait loading patterns during daily	Both devices detect dangerous plantar loading and generate

Feature / Function	Predicate: SmartStep System (K060150)	NeuroFoot (Proposed Device)	Substantial Equivalence
	alert the user/therapist when the weight exceeds the pre-selected value	ambulation; alert user and clinician when sustained loading or composite risk score exceeds clinically validated thresholds	user-facing alerts. NeuroFoot extends monitoring from rehabilitation to daily wear and adds gait-based risk inputs
Indications for Use	Adults in rehabilitation requiring partial weight-bearing monitoring under physician/physiotherapist supervision (Prescription Use)	Adults with diabetic peripheral neuropathy are at elevated risk for plantar ulceration under clinician supervision (Prescription Use)	Both target patients require external monitoring of plantar stress due to the inability to self-regulate loading. Both require clinical oversight
Sensor Technology	Flexible resistive pressure-sensing insole	Flexible resistive pressure-sensing insole (16-zone FSR array via CD74HC4067 multiplexer)	Same fundamental sensing modality (resistive force sensing in insole form factor)
Pressure Measurement	Plantar pressure magnitude; threshold-based overload detection	16-zone plantar pressure at 50 Hz; peak pressure, Pressure-Time Integral (PTI), and zone-specific sustained-load detection	NeuroFoot provides higher spatial resolution and adds PTI as a clinically validated biomarker. Core function is the same
Alert / Warning System	Audible and/or visual alarm when the weight exceeds the pre-set threshold	Multi-level alert (OK / WARN / CRITICAL) via smartphone app with haptic, audible, and visual notifications; composite risk score	Both provide real-time overload warnings. NeuroFoot adds graded severity and mobile delivery
Environmental Sensing	Not included	Onboard HS300x sensor provides ambient temperature (±0.25°C) and humidity (±3.5% RH) for environmental context logging	Supplementary environmental data does not change the primary intended use
Gait / IMU	Not included	BMI270 6-axis IMU (accel ±4g, gyro ±2000 dps) at 50 Hz; step count, cadence, stance duration, lateral balance	Supplementary motion data enhances pressure context; it does not change the fundamental device purpose
Wireless Communication	The Control Unit communicates with the PC Software wirelessly	Bluetooth Low Energy (BLE 5.0) to iOS companion app	Both use wireless data transmission to an external display
Form Factor	Flexible insole + belt-worn Control Unit + PC Software	Flexible sandal with embedded FSR array and integrated microcontroller (Arduino Nano 33 BLE Sense Rev2) + iOS smartphone app	Both are wearable plantar-sensing systems with external processing and display
Data Display	PC-based software interface; real-time pressure visualization	iOS app with real-time plantar pressure heatmap, zone bars,	Both provide real-time visual feedback of plantar data

Feature / Function	Predicate: SmartStep System (K060150)	NeuroFoot (Proposed Device)	Substantial Equivalence
		gait metrics, environmental readings, and event log	
Clinical Setting	Rehabilitation clinic and supervised home use	Continuous daily wear with remote clinician access to trend data	NeuroFoot extends monitoring beyond rehabilitation. No new safety risks as the device remains non-invasive and prescription-supervised
Safety Standards	IEC 60601-1, IEC 60601-1-2, AAMI/ISO 14971-1	Target compliance: IEC 60601-1, IEC 60601-1-2, ISO 14971, IEC 62304 (software lifecycle)	Same core safety standards. NeuroFoot adds IEC 62304 for mobile software component
Patient Contact	Non-invasive; insole worn inside the shoe	Non-invasive; sandal worn on foot, microcontroller housed between sandal layers	Identical patient contact profile (non-invasive external wear)

1.2. Design Outputs Supporting Production

1.2.a. Technical Drawings, BOM, and Component Specifications

The bill of materials captures every component in the NeuroFoot prototype, along with its manufacturer part number, vendor, quantity, and the key specification that drove its selection. Table 6 presents the full BOM. Each component was selected to satisfy a specific design input requirement identified in Section 1.1, and the rationale for choosing one part over alternatives is discussed below the table.

Table 8. NeuroFoot Bill of Materials

Component	Part Number / Model	Vendor	Qty	Key Specification
Microcontroller	Arduino Nano 33 BLE Sense Rev2 (ABX00070)	Arduino.cc	1	nRF52840 SoC; BLE 5.0; 12-bit ADC; onboard BMI270 IMU, HS300x temp/RH
Analog Multiplexer	CD74HC4067 (16-ch)	Texas Instruments	1	16:1 analog mux; 3.3V logic; ~100 ns switching; single analog output
FSR Insole Sensor	16 Pressure Sensors	Walfront (Amazon)	1	16 discrete zones; 1 kΩ–10 MΩ range; covers heel, midfoot, forefoot
Pull-down Resistors	10 kΩ, 1/4 W, ±5%	Generic (Amazon)	16	Voltage divider with FSR; ~930-count usable ADC range at 3.3V
Breadboard	Half-size solderless breadboard	Generic	1	400 tie points; prototyping platform for mux + resistor network
Sandal Base	EVA foam sandal (2-layer)	Generic	1	Midsole cavity houses Arduino + breakout; two layers sandwich electronics

Component	Part Number / Model	Vendor	Qty	Key Specification
USB Cable	Micro-USB to USB-A	Generic	1	Power and serial programming; 5V/500 mA from host

The Arduino Nano 33 BLE Sense Rev2 was selected as the microcontroller because it integrates the nRF52840 SoC with BLE 5.0 radio, a BMI270 6-axis IMU, and an HS300x temperature and humidity sensor on a single 45 × 18 mm board. Using a module with onboard sensors eliminates the need for external breakout boards, reduces wiring complexity, and keeps the overall footprint small enough to embed inside a sandal midsole. Alternative microcontrollers such as the ESP32 were considered, but the ESP32 lacks an onboard IMU and environmental sensor, which would have required two additional breakout boards and significantly increased the assembly size.

The CD74HC4067 16-channel analog multiplexer was chosen to route the 16 FSR zones through a single analog input pin on the Arduino. The Nano 33 BLE Rev2 has only 8 analog inputs, so a multiplexer is necessary to read all 16 zones. The CD74HC4067 was selected over the CD4051 (8 channels) because it handles all 16 zones without requiring a second multiplexer chip, and its channel switching time of approximately 100 nanoseconds is well within the 100-microsecond settle time used in firmware. The device operates at 3.3V logic, which matches the Nano 33 BLE's I/O voltage without level shifting.

An inhouse developed 16-zone FSR insole sensor was used. It provides 16 discrete sensing zones covering the heel, midfoot, and forefoot regions. Each zone exhibits a resistance range of approximately 1 kΩ under full load to 10 MΩ unloaded, which produces a usable ADC range when paired with a 10 kΩ pull-down resistor in a voltage divider configuration. Figure 1 illustrates the PCB layout and locations of the resistors, capacitors, and pad locations for future reference.

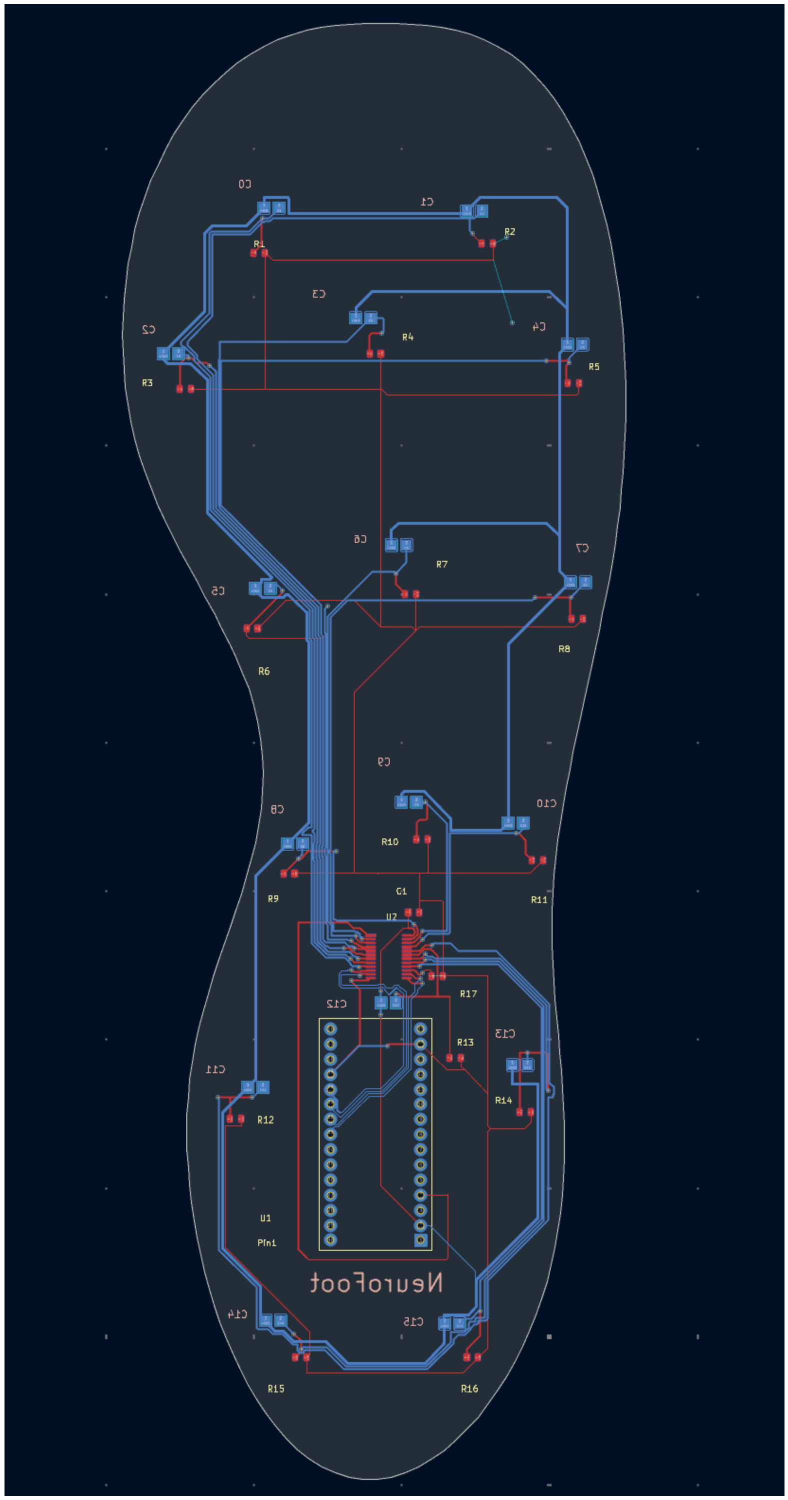

Figure 1: Neurofoot PCB schematic and layout

Vendor selection for the three critical components was guided by four criteria: technical fit with the design input requirements, supply chain reliability, documentation quality, and cost at prototype quantities. For each component, at least one alternative was evaluated before a final selection was made.

Arduino.cc was selected as the microcontroller vendor because it is the sole authorized manufacturer of the Nano 33 BLE Sense Rev2 (ABX00070). Third-party clones of the Nano 33 BLE exist, but they do not include the onboard BMI270 IMU or HS300x environmental sensor, which would defeat the integration advantage that drove the board selection in the first place. Arduino.cc provides complete datasheets, pinout diagrams, and an actively maintained board support package through the Arduino IDE, which reduces firmware development risk. The board is available directly from Arduino.cc for approximately $35 at single-unit quantities, with typical lead times under one week. For production volumes, Arduino offers the Nano 33 BLE Sense Rev2 through authorized distributors including Mouser, Digi-Key, and RS Components, all of which provide quantity pricing and full traceability documentation suitable for ISO 13485 purchasing controls. Vendor quality certification note: Arduino.cc and its authorized distributors (Mouser, Digi-Key, RS Components) operate under ISO 9001 quality management systems; none currently hold ISO 13485 certification. Under 21 CFR 820.50 (Purchasing Controls), the manufacturer is responsible for establishing supplier qualification requirements commensurate with risk. For production, the purchasing team must conduct a supplier evaluation confirming: (1) the vendor's ISO 9001 certificate is current and in-scope for the relevant product category; (2) component traceability records (lot number, date code, certificate of conformance) are available per order; and (3) a supplier agreement is in place defining quality and notification requirements. ISO 9001 certification is an acceptable baseline for a component supplier provided these additional medical-device-specific controls are documented in the Approved Supplier List (ASL). A formal gap assessment against ISO 13485 Section 7.4 requirements must be completed and filed before transitioning from prototype to production procurement.

Texas Instruments was selected as the multiplexer vendor because the CD74HC4067 is a well-documented, widely available CMOS analog multiplexer with decades of production history. The primary alternative considered was the Analog Devices ADG706, which offers lower on-resistance but at roughly three times the unit cost and with longer lead times at prototype quantities. Since the FSR voltage divider circuit is not sensitive to on-resistance variations in the range of 50-100 ohms, the CD74HC4067 meets all electrical requirements at lower cost. Texas Instruments provides a complete datasheet with timing diagrams, application notes, and SPICE models, which simplified the circuit design and will support PCB layout validation. The device is stocked at all major distributors and is classified as a long-lifecycle product, reducing the risk of supply disruption during the transition from prototype to production.

1.2.b. Acceptance Criteria and Component Verification

Each component in the NeuroFoot system was validated individually before integration, and the integrated system was validated as a whole before use in walking trials. The acceptance criteria were derived directly from the MVP specifications established in the project proposal and from the sensor datasheets. A component was accepted only if it met the defined pass criteria under the specified test method. Components that failed acceptance testing were either replaced, reconfigured, or excluded from the final design.

Table 9 defines the acceptance criteria for each subsystem. The tests are ordered from individual component validation to full system integration, reflecting the verification sequence used during prototyping.

Table 9. Component and System Acceptance Criteria

Component	Acceptance Test	Pass Criteria	Test Method
Arduino Nano 33 BLE Sense Rev2	Power-on self-test	Serial output confirms board ID, BLE MAC address, and firmware version within 5 seconds of reset	Connect via USB, open Serial Monitor at 115200 baud, verify output after pressing RESET
CD74HC4067 Multiplexer	Channel isolation test	Each channel reads <5 ADC counts when unloaded (with 10k pull-down), and all other channels remain below 5 ADC counts during single-channel stimulation	Apply 3.3V to one channel input through a 10k resistor, scan all 16 channels via MuxPressureTest sketch MODE 1, verify only the stimulated channel shows an elevated reading
FSR Array (16 zones)	Response linearity check	Monotonically increasing ADC output across 0, 5, 10, 20, 40 kg applied loads; R-squared greater than 0.90 for linear fit; all 16 channels respond independently	Mount insole on rigid surface, apply calibrated weights to individual zones, record ADC values through MuxPressureTest MODE 1, plot and compute correlation
BMI270 IMU (Accelerometer)	Six-position static calibration	Each axis reads ±1g (±0.05g tolerance) when aligned with gravity; near 0g on orthogonal axes	Place the device flat, then rotate to each of 6 orientations (+X, -X, +Y, -Y, +Z, -Z up), record 100 samples per orientation, and compute the mean per axis
BMI270 IMU (Gyroscope)	Zero-rate bias test	All three axes read below ±0.5 dps when stationary for 30 seconds	Place the device on a stable surface, collect 3000 samples at 100 Hz, and compute the mean angular velocity per axis
HS300x (Temperature)	Ambient reference comparison	Reads within ±0.5°C of a calibrated reference thermometer at room temperature (20–25°C)	Place the device and reference thermometer side by side in stable ambient conditions for 5

Component	Acceptance Test	Pass Criteria	Test Method
			minutes, and compare readings
HS300x (Humidity)	Ambient reference comparison	Reads within ±5% RH of a calibrated reference hygrometer at ambient conditions	Place the device and reference hygrometer side by side, allow 5 minutes of stabilization, and compare readings
BLE Communication	Connection and throughput test	Device advertises within 3 seconds of power-on, iOS app connects within 10 seconds, all four characteristics (pressure, IMU, env, status) receive data at expected rates (pressure/IMU at 50 Hz, env at 0.5 Hz)	Power on the device, open the NeuroFoot iOS app, initiate connection, verify all data fields populate on the Monitor screen within 15 seconds
Alert System	Threshold trigger test	WARN alert triggers within 1 second when any sensor exceeds 600 ADC; CRITICAL alert triggers within 1 second when any sensor exceeds 850 ADC; alert clears within 2 seconds of pressure removal	Apply known pressure to the single zone exceeding each threshold, measure time from threshold crossing to alert appearance on the app, release, and measure clear time
Integrated System	Walk test	During a 2-minute walking trial, the system continuously displays updating pressure heatmap, gait metrics increment with each step, no data dropouts or BLE disconnections	Wear sandals, walk at a normal pace for 2 minutes while a second person monitors the iOS app, verify continuous data flow, and correct the step count

Integration verification was performed in two stages. In the first stage, the Arduino was connected to the multiplexer and FSR array on a breadboard, and the MuxPressureTest sketch was used to confirm that all 16 channels produced independent, noise-free readings with 10k pull-down resistors installed. The NeuroFoot_BLE sketch was then uploaded, and BLE connectivity was verified by confirming that the iOS app received live data on all four characteristics (pressure, IMU, environment, status) simultaneously. In the second stage, the breadboard prototype was transferred to the sandal assembly. The Arduino board was secured between the two structural layers of the sandal, and the full system walk test (Table 9, final row) was performed to verify that the mechanical assembly did not introduce signal degradation, loose connections, or BLE range issues.

One component was rejected during acceptance testing. The BMM150 magnetometer, which is integrated on the Nano 33 BLE Sense Rev2 board, consistently returned overflow values (-32768) on all three axes. An I2C bus scan confirmed that the BMM150 is not directly addressable on Wire1 at its expected address (0x10) because it is routed through the BMI270's internal auxiliary interface, which the Arduino_BMI270_BMM150 library v1.2.3 does not properly support on the Rev2 hardware. After six debugging approaches failed to resolve the issue, the magnetometer was excluded from the final design. This exclusion does not affect the device's intended use, as the accelerometer and gyroscope provide sufficient motion and gait data for the clinical application.

1.2.c. Traceability and PCB Translation

Design traceability links each functional requirement from the project proposal to a specific BOM component, documents the breadboard validation result for that component, and identifies the design considerations that must carry forward into a PCB layout. This traceability ensures that decisions made during prototyping are systematically preserved as the design moves from breadboard to production PCB, and that lessons learned from testing are not lost when a different team member performs the PCB layout. Table 10 presents the traceability matrix.

Table 10. Design Input Traceability Matrix: Requirements to BOM to PCB

Design Input (Requirement)	BOM Component	Breadboard Validation Result	PCB Design Consideration
16-zone plantar pressure at 50 Hz, 0–400 kPa	Walfront 16-zone FSR insole + CD74HC4067 mux + 10k pull-down resistors (x16)	Confirmed monotonic ADC response across load range; 100 microsecond mux settle time sufficient; 10k pull-downs eliminate 60 Hz floating-input noise	Route 16 mux channel traces with matched lengths; place 10k pull-down resistor array adjacent to mux; include 100nF bypass cap on mux VCC
6-axis IMU at 50 Hz	BMI270 (onboard Arduino Nano 33 BLE Sense Rev2)	Accelerometer and gyroscope verified via six-position static test and zero-rate bias test; magnetometer abandoned (BMM150 not accessible on Rev2 Wire1 bus)	No external routing needed (onboard); ensure IMU ground plane is uninterrupted beneath the Arduino module footprint; do NOT call Wire.begin() in firmware as it corrupts the internal I2C bus
Ambient temperature and humidity logging	HS300x (onboard Arduino Nano 33 BLE Sense Rev2)	Confirmed ±0.5°C against reference thermometer; first-boot zero-read resolved with retry loop in firmware	No external routing needed (onboard); ensure adequate ventilation holes in sandal enclosure layer near the sensor area; add firmware retry loop for first-boot initialization
BLE wireless at <1 second alert latency	nRF52840 BLE radio (onboard Arduino Nano 33 BLE Sense Rev2)	Measured <200 ms from threshold crossing to iOS app alert display; connection established within 10 seconds; service UUID filter reliable across iOS devices	Keep the BLE antenna area (top edge of Nano module) free from ground fill and copper traces; do not place components within 5 mm of the antenna; orient the antenna toward the sandal exterior

Design Input (Requirement)	BOM Component	Breadboard Validation Result	PCB Design Consideration
Alert thresholds: WARN >600 ADC, CRITICAL >850 ADC	Firmware logic on nRF52840	Confirmed correct threshold triggering in bench test with controlled loads; alert clears within 2 seconds	Thresholds stored as firmware constants; consider adding storage for clinician-adjustable thresholds in the production version

Components to Avoid

Prototype testing identified several components and configurations that should be avoided in the production design. First, 100k ohm pull-down resistors must not be used with the FSR voltage divider circuit. The team initially used 100k pull-downs, which compressed the useful ADC range to approximately 270 counts (410–682) for the 50K–150K ohm sensor range, severely limiting measurement resolution. Replacing them with 10k pull-downs expanded the usable range to approximately 930 counts, providing adequate resolution for clinical pressure differentiation. Second, piezoelectric sensors should not be used as a substitute for force-sensitive resistors in this application. Early testing with piezoelectric elements showed that they generate their own voltage on impact but require a bleed resistor (1M ohm) to drain the charge, are highly susceptible to 60 Hz mains hum without shielding, and do not provide a sustained reading under static load. Since the clinical use case requires detecting sustained pressure over time, not just impact events, piezos are fundamentally unsuitable. Third, the BMM150 magnetometer should not be relied upon in any Rev2-based design.

As documented in the acceptance testing section, the sensor is not accessible through the standard I2C interface on this board revision. Fourth, calling Wire.begin() in firmware must be avoided on the Nano 33 BLE Sense Rev2, as it corrupts the onboard sensors' I2C bus. The HS300x and BMI270 communicate internally over Wire1, and initializing Wire (the external bus) at the wrong time causes sensor read failures. Fifth, the APDS9960 color/proximity sensor must be initialized last in the firmware setup sequence. If initialized before other I2C devices, it causes bus conflicts that prevent the IMU and temperature sensors from responding.

1.2.d. Manufacturing Process Flow

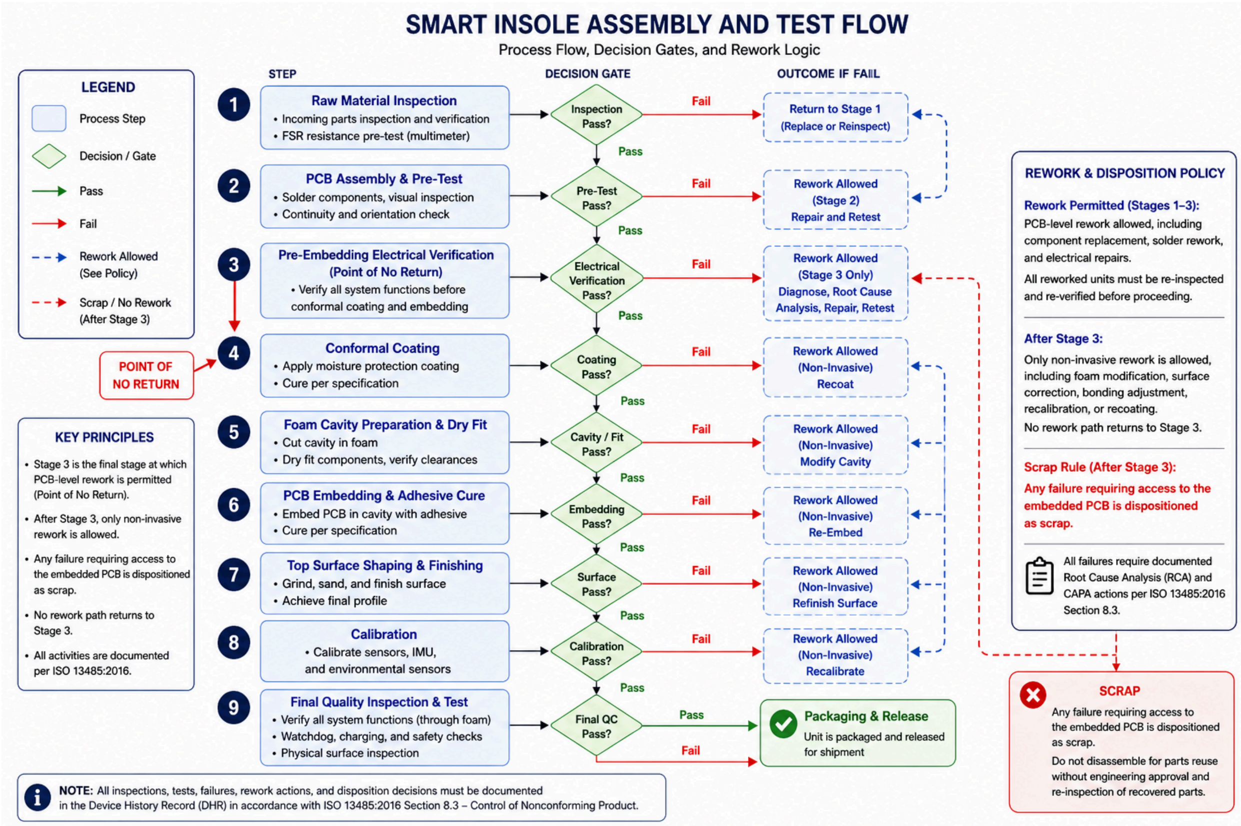

Figure 2. NeuroFoot Assembly and Test Flow. A stage-gated manufacturing and verification process with decision gates at each step. Full PCB-level rework is permitted prior to Stage 3 (pre-embedding electrical verification). After Stage 3, only non-invasive rework is allowed. Any failure requiring access to the embedded PCB is dispositioned as scrap. All failures require documented root cause analysis and CAPA actions in accordance with ISO 13485:2016 Section 8.3.

The production constraints reflect the fact that the NeuroFoot prototype is assembled by hand, and that several steps in the assembly process are not easily repeatable without standardization. The rigid FR4 PCB with fixed component positions and documented sensor coordinates ensures that every unit is assembled identically regardless of operator. Standardized foam cavity dimensions and controlled adhesive application eliminate unit-to-unit variation in sensor depth and lateral positioning. Small, medium, and large size variants follow the same assembly sequence; only the PCB and sandal dimensions change, not the process steps or inspection criteria. All electrical test checkpoints are defined with explicit pass/fail criteria so that any trained assembler can make a go/no-go decision without engineering judgment. Pre-solder FSR testing eliminates the risk of building defective sensors into completed assemblies where rework is impractical.

The full assembly process is organized into ten stages, each with defined inputs, outputs, and decision gates. Figure 2 presents the assembly flowchart from start to deployment, showing the sequence of process steps, inspection decision gates, critical gates (shown in green), calibration steps, and rework loops. Rework loops return to the earliest stage that could have caused the failure, and no rework loop passes back through Stage 3. Any failure requiring PCB access after embedding results in a scrapped unit.

Stage 1 is component procurement and inspection. Before assembly begins, all components are verified against the bill of materials, and no assembly proceeds with missing or suspect parts. The PCB is checked for visible trace damage, lifted pads, or contamination, and the footprint is confirmed against the current revision. All 16 FSR sensors are physically inspected, and any sensor with visible creases, delamination, or damage to the sensing element is rejected. Each FSR undergoes a pre-solder test using a multimeter set to resistance mode: the sensor must read near-infinite resistance with no load applied and must show a measurable drop in resistance when pressed firmly by hand. Any sensor that fails either check is replaced before it can be soldered into the assembly. The Arduino Nano 33 BLE Sense Rev2 is confirmed to power on and be BLE-discoverable, the CD74HC4067 multiplexer is confirmed for correct part number and pin count, the LiPo battery is inspected for charge level, swelling, and connector compatibility, and the foam sandal is confirmed for correct size variant, matching durometer, and freedom from material defects.

Stage 2 is PCB population and soldering. Components are soldered in a fixed order to avoid thermal interference and access issues: the CD74HC4067 multiplexer first, then the Arduino Nano 33 BLE Sense Rev2, followed by the power connector, charging circuit, and battery leads, then the alert output components (buzzer and/or vibration motor), and finally all 16 FSR sensors in order of anatomical zone number. Each FSR must sit flat against the PCB surface with no lifting at edges. Stencil and reflow oven processing is preferred where available, but hand soldering is acceptable with mandatory joint inspection after each component group. After all components are placed, every solder joint is visually inspected. Joints must be shiny, concave fillets with no bridging, cold joints, or lifted pads. Any non-conforming joint is reworked before proceeding.

Stage 3 is pre-embedding electrical verification and represents the final stage at which PCB-level rework is permitted. This stage must be completed and passed before any conformal coating or foam embedding begins, because once the device is embedded, rework requiring access to the PCB is no longer possible. The assembled PCB is powered on with the LiPo battery, and diagnostic firmware is uploaded to the Arduino. Each of the 16 multiplexer channels is cycled to verify that every channel produces a distinct and responsive signal when its corresponding FSR is actuated. Cross-talk testing is performed to verify that actuation of one sensor does not produce measurable signal changes in adjacent channels. The sampling rate is verified against the 10 to 50 Hz specification by logging timestamps of consecutive readings. The alert output is verified by simulating a predefined engineering threshold exceedance in firmware to confirm that the buzzer or vibration motor activates correctly. This alert indicates a sensor-derived event only and does not represent a clinical determination. BLE discoverability and data transmission are verified. The BMI270 IMU is confirmed to initialize and return accelerometer and gyroscope data on all six axes, and the HS300x temperature and humidity sensor is confirmed to initialize, with up to three retries permitted on cold boot failure before flagging a component fault. The pass criteria require that all system functions operate as intended, including sensor responsiveness, absence of cross-talk, sampling rate compliance, alert activation, BLE communication, and IMU functionality. If a single sensor channel is non-responsive, the system shall flag the channel as inactive. The unit may proceed only if degraded operation is acceptable and documented; otherwise, the unit must be reworked. Any failure identified at this stage requires root cause analysis and correction before proceeding. Rework is permitted within Stage 3 until all requirements are met.

The nonconformance disposition rule, per ISO 13485 Section 8.3, distinguishes between reworkable and non-reworkable failures. At Stages 1 through 3, rework is permitted. Any solder joint, component orientation, or electrical continuity failure identified prior to embedding may be corrected by a trained assembler, and the reworked joint or component must be re-inspected and documented before proceeding. After completion of Stage 3, rework is limited to non-invasive processes such as foam modification, surface correction, bonding adjustment, or calibration. No rework path may return to Stage 3. Any unit that fails after Stage 3 and requires access to the embedded PCB, including failures identified during Stage 7 surface inspection or Stage 9 final quality control, must be quarantined, tagged as nonconforming, and dispositioned as scrap. A nonconformance record must be opened for each scrapped unit, documenting the failure mode, the stage at which it was detected, and the root cause where determinable. Scrapped units must not be disassembled for parts reuse without explicit engineering approval and re-inspection of recovered components.

Stage 4 is conformal coating. The active sensing areas on each FSR are masked with tape or dot masks, as are the USB-C or charging port and any exposed connector pins. Conformal coat is applied to all PCB traces, solder joints, and component bodies using brush or spray application, and the coating is allowed to cure fully per the manufacturer specification before handling. After cure, the masks are removed and coating coverage is inspected under UV light if available.

Stage 5 is foam cavity preparation. The cavity outline is marked on the foam sandal base using the size-matched PCB template, aligned to anatomical reference marks on the sandal. The cavity is cut to the specified depth so that the PCB seats with sensor surfaces facing upward at the correct height relative to the sandal top surface. A dry fit is performed without adhesive to confirm that the PCB seats flush, does not rock, and that sensor positions align with the anatomical zone markers on the template. A separate channel is cut for the battery and wiring if the battery is housed separately from the main PCB.

Stage 6 is PCB embedding. Epoxy adhesive is applied evenly to the cavity floor, avoiding the sensor active areas. The PCB is seated into the cavity and aligned to the anatomical zone template before the adhesive sets. Even downward pressure is applied across the full PCB surface, with clamping or weighting as needed during cure. After full adhesive cure, the PCB is confirmed to remain stationary when lateral force is applied.

Stage 7 is foam resealing. Waterproof adhesive is applied evenly to the foam lid contact surface, and the top layer is pressed firmly over the PCB cavity with even pressure applied across the full surface during cure to prevent uneven foam thickness above the sensors. After cure, the top surface is inspected: it must be completely flush with no ridges, seams, or bumps detectable by hand. Any surface irregularity above a component location is a patient safety concern and must be reworked before proceeding.

Stage 8 is IMU calibration. The assembled sandal is placed on a flat, level surface and powered on. The accelerometer static offset is recorded on all three axes with no movement, and the offset values are stored in firmware for automatic subtraction during operation. The gyroscope zero-rate offset is recorded at rest and stored as a correction factor; the gyroscope must be re-zeroed at the start of each wear session. The physical orientation of the Arduino Nano on the PCB relative to foot axes (heel-to-toe, medial-lateral, and vertical) is documented, and a fixed rotation matrix is applied in firmware so that IMU output is always in anatomically meaningful coordinates.

Stage 9 is the final quality inspection and test and serves as the last gate before packaging. A physical inspection is performed by running a hand over the full top surface to verify that no ridges, bumps, or hard spots are detectable through the foam, ensuring user safety and comfort. The fully assembled device is powered on, and all sensor channels are evaluated to confirm that pressure inputs are detectable through the foam layer. The alert output is verified by simulating a predefined engineering threshold exceedance to confirm that the buzzer or vibration motor activates correctly. This alert indicates a sensor-derived event only and does not constitute a clinical diagnosis. BLE connectivity and data transmission to the companion system are verified, the IMU is confirmed to return valid data post-embedding, the watchdog and heartbeat fault detection system are confirmed active, and battery charging is verified via USB-C or magnetic connector. The pass criteria require that all system functions operate as intended, including sensor responsiveness, alert activation, wireless communication, IMU functionality, fault detection, charging capability, and a smooth surface profile. If a failure is identified, the unit is returned to the appropriate prior stage for root cause analysis and non-invasive rework where possible. Any failure that requires access to the embedded PCB is dispositioned as scrap. No device is packaged without having fully passed this final inspection.

Stage 10 is packaging and labeling. Each device is labeled with its size variant, serial number, and date of assembly. Charging instructions, an alert guide, and a recalibration schedule are included in the packaging, and the device is packaged for storage or deployment.

1.2.e. Prototype Verification and Testing

Prototype verification was performed in two stages that mirror the assembly flowchart presented in Section 1.2.d. The first stage validated the electronics on a breadboard before any foam embedding, corresponding to Stages 1 through 3 of the assembly process. The second stage validated the fully assembled sandal as an integrated system, corresponding to Stages 6 through 9. This two-stage approach ensures that electrical failures are caught while rework is still possible (before the Stage 3 point of no return) and that mechanical assembly does not introduce signal degradation, loose connections, or BLE range issues.

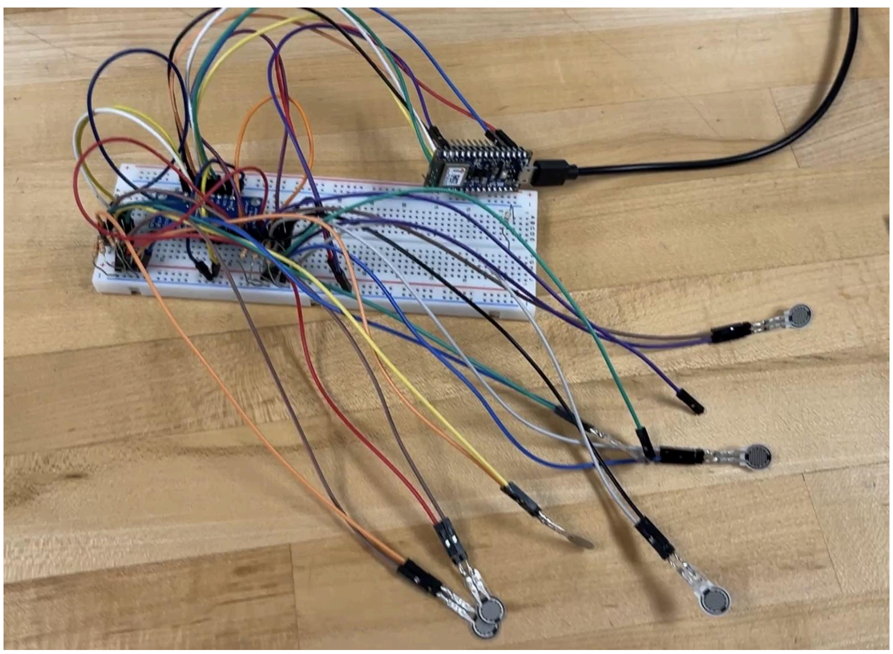

Figure 3. Breadboard validation of the NeuroFoot electronics (Stage 3 equivalent). The Arduino Nano 33 BLE Sense Rev2, CD74HC4067 multiplexer, 16 pull-down resistors, and FSR sensor leads are wired on a solderless breadboard. The companion iOS app (shown on phone at left) receives live pressure, IMU, environmental, and status data over BLE.

This configuration was used to confirm that all 16 multiplexer channels produced independent, noise-free readings with 10 kΩ pull-down resistors, that BLE connectivity was stable, and that the alert threshold logic triggered correctly. The MuxPressureTest diagnostic sketch was used to cycle through all channels and verify that no cross-talk existed between adjacent zones. This breadboard stage also revealed the BMM150 magnetometer failure: the sensor consistently returned overflow values on all three axes because its I2C address (0x10) is not directly accessible on the Rev2 hardware, being routed through the BMI270's internal auxiliary interface. After six debugging approaches failed to resolve the issue, the magnetometer was excluded from the final design. This exclusion does not affect the device's intended use, as the accelerometer and gyroscope provide sufficient motion and gait data for the clinical application. Nine of ten components passed acceptance testing at this stage.

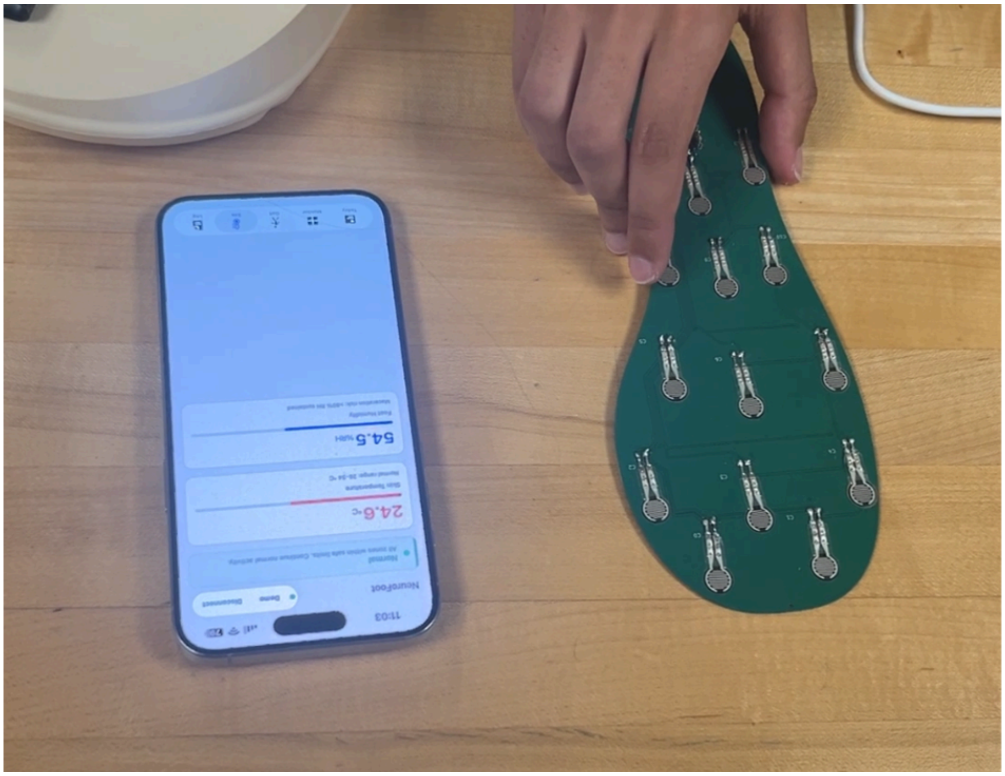

Figure 4. FSR sensor response verification on the custom insole PCB. An operator presses individual sensor zones on the populated PCB (green, foot-shaped board) while monitoring real-time ADC readings on the companion app.

This step corresponds to the channel-by-channel verification in Stage 3 of the assembly flowchart and to the FSR array response linearity check defined in the acceptance criteria (Table 6, Section 1.2.b). Each of the 16 zones was tested for monotonically increasing ADC output across applied loads, and the R-squared value for linear fit was confirmed to be greater than 0.90 for all channels. This image also illustrates the custom insole construction: the individual MakerHawk RP-C thin film pressure sensors are soldered at fixed positions across the board rather than using a pre-made sensor array, which allows precise placement at the anatomically relevant zones identified in the stakeholder analysis.

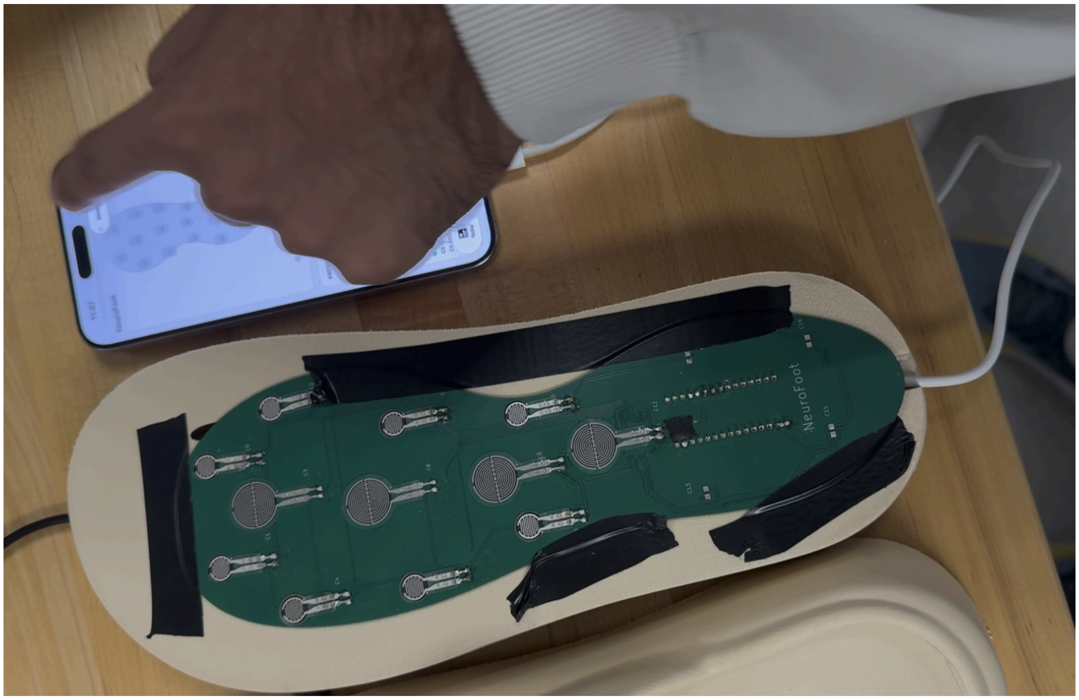

Figure 5. Completed NeuroFoot sandal assembly during the integrated walk test (Stage 9). The PCB is embedded between the two EVA foam layers, the foam has been resealed, and the device is powered on and transmitting live data to the companion iOS app via BLE.

This configuration represents the final acceptance test defined in the assembly flowchart: the system must continuously update the pressure heatmap and gait metrics with each step, with no data dropouts or BLE disconnections over a two-minute walking trial at normal pace. The walk test confirms that the mechanical assembly process (cavity cutting, PCB bonding, foam resealing, and conformal coating) did not introduce signal degradation or sensor misalignment relative to the pre-embedding baseline recorded during Stage 3.

Integration verification confirmed that the transition from breadboard to embedded sandal preserved full system functionality. All pressure channels, IMU axes, environmental sensors, BLE link, and alert thresholds were confirmed operational in the final assembled form factor. The only component excluded from the final design was the BMM150 magnetometer, which was rejected during breadboard validation as described above. A detailed record of each component's acceptance test, pass criteria, and test method is provided in Table 6 of Section 1.2.b.

2. Design for Failure Mode Analysis (FMEA)

This section examines how the device could fail, assesses the clinical severity and likelihood of each failure, and defines design controls that reduce risk to acceptable levels. The approach follows ISO 13485 Section 7.1 and 7.3.7. The three subsections cover failure mode identification, risk scoring and prioritization, and mitigation strategies.

2.1. Identification of Failure Modes

Before risks can be scored or mitigated, they must first be identified. This subsection documents the failure modes discovered through prototype testing and those anticipated based on the device's mechanical and electrical design. Each failure mode is described in terms of its mechanism and its potential clinical consequence for the neuropathic patient population.

2.1.a. Potential Failure Modes

The following failure modes were identified through a systematic review of the NeuroFoot prototype, drawing on failures observed during breadboard and sandal assembly testing, analysis of the physical construction and wiring, and consideration of the clinical use environment. Five failure modes are identified across four categories: mechanical wear, environmental exposure, wireless communication, and user error. Table 11 summarizes the identified failure modes; each is described in detail below.

Table 11. Summary of Identified Failure Modes

Failure Mode	Category	Subsystem	Source
FSR signal degradation from adhesive delamination	Mechanical Wear	Pressure Sensing	Anticipated
60 Hz electromagnetic interference	Environmental	Pressure Sensing (ADC)	Observed in prototype
Moisture ingress causing short or corrosion	Environmental	Electronics / Sandal Assembly	Anticipated
BLE disconnection during ambulatory use	Wireless Communication	BLE Radio / iOS App	Anticipated
User fails to charge the device before daily use	User Error	Power Management	Anticipated

The FSR insole is bonded to the sandal sole using adhesive. Over repeated flexion cycles during walking, the adhesive layer may delaminate, creating an air gap that reduces mechanical coupling between the foot and the sensing element. This causes the system to underreport actual plantar loading. Because the patient has a loss of protective sensation, they cannot independently verify whether the device is accurately reflecting their foot pressure, and underreported pressure may result in missed alerts during sustained overload. Another risk could also be determined when after extended usage, the foam above the pressure sensors changes in material qualities, which leads to elevated readings and excessive signaling when not clinically relevant.

The multiplexer's analog signal line acts as an antenna for electromagnetic interference from AC mains wiring. This was directly observed during prototype testing: with no pull-down resistors installed, the ADC readings fluctuated across the full 0–1023 range at 60 Hz, and in the 16-channel scan, the noise peak traveled across channels because each sample was at a slightly different phase in the cycle. While 10k pull-downs attenuate this under normal conditions, environments with strong EMI could still cause false threshold crossings, leading to alert fatigue.

The Arduino and multiplexer are housed between two sandal layers, but the current prototype has no sealed or conformal-coated enclosure. Rain, puddles, or accumulated perspiration may reach the PCB and connector interfaces. Moisture on the multiplexer or connector pins can cause intermittent short circuits between adjacent channels or oxidize contacts over time. Furthermore, due to the multiplexer and arduino being pressure sensitive components, integration of them within the sole leads to potential in shoring signals due to excessive force on components which were not designed for those mechanical exposures.

The BLE link between the Arduino and the iOS app may drop due to RF interference from other 2.4 GHz devices, exceeding the effective range, antenna obstruction by the sandal layers, or the iOS suspending the app. During a disconnection, no alerts are sent to the patient. The firmware continues local monitoring and illuminates the red LED during a CRITICAL event, but this indicator on the ankle-mounted board may not be visible while walking. These signals also need to be encrypted end to end to comply with HIPAA regulations.

If the patient forgets to charge overnight or misses the low-battery notification, the device may shut down during use, and all monitoring ceases. Patients with diabetic peripheral neuropathy may have cognitive comorbidities or limited smartphone familiarity, increasing the likelihood of a missed warning. The current prototype delivers the low-battery warning only through the iOS app, with no independent audible alarm. Notifications on connecting the device are propagated every day at 9AM to ensure the device will be used during all walking durations.

These five failure modes are anticipated based on the prototype's design and are carried forward into the risk analysis in Section 2.2 and the mitigation strategies in Section 2.3.

2.2. Risk Analysis & Prioritization

2.2.a. Risk Scoring (Severity, Occurrence, Detection)

Risk Priority Number (RPN) is calculated as the product of three independent scores: Severity (S), Occurrence (O), and Detection (D). Each dimension is rated on a scale of 1 to 10. Scores are assigned based on data available at the Preliminary Design Review stage, which includes component manufacturer datasheets, peer-reviewed biomechanical literature, and engineering judgment where empirical data is not yet available. All score rationales are documented in the FMEA table, and scores will be revised following Design Verification testing.

Severity describes the worst-case consequence of a failure mode reaching the patient, assuming the failure is not detected or corrected. Because the NeuroFoot is intended for adults with diabetic peripheral neuropathy and loss of protective sensation (IWGDF Risk 2 to 3), the severity scale must reflect the clinical reality that these patients cannot independently detect many device failures. A minor failure (score 1 to 3) slightly degrades device function without any patient safety impact, and the patient or clinician would notice and self-correct without consequence. A moderate failure (score 4 to 6) reduces monitoring accuracy or affects alert frequency, potentially causing user inconvenience or alert fatigue, but no direct tissue injury is expected. A serious failure (score 7 to 8) causes a core alert or monitoring function to fail, exposing the patient to sustained plantar mechanical overload without warning and producing a clinically meaningful elevation in diabetic foot ulcer risk. A critical failure (score 9 to 10) either causes direct physical injury, such as electrical shock or skin damage from hardware failure, or results in complete silent failure while the patient believes the device is still operational. Table 12 summarizes the full severity scale with NeuroFoot-specific clinical context.

Table 12: Severity Scale and NeuroFoot Clinical Context

Score Range	Description	NeuroFoot Clinical Context
1 to 3	Minor	Device function slightly degraded; no patient safety impact; patient or clinician would notice and self-correct without consequence
4 to 6	Moderate	Monitoring accuracy reduced or alert frequency affected; potential for user inconvenience or alert fatigue; no direct tissue injury expected
7 to 8	Serious	Core alert or monitoring function fails; patient is exposed to sustained plantar mechanical overload without warning; clinically meaningful DFU risk elevation
9 to 10	Critical	Device causes direct physical injury (e.g., electrical shock, skin damage from hardware failure); or complete silent failure with patient believing device is operational

Occurrence reflects the estimated probability that the identified failure cause will produce the failure mode during the expected device lifetime, based on available evidence at the PDR stage. The scale ranges from remote (score 1 to 2), representing fewer than 1 in 10,000 operating cycles where the component-level specification provides very high reliability assurance, through low (score 3 to 4) at approximately 1 in 1,000 cycles where the component meets specification but failure is possible under normal use, to moderate (score 5 to 6) at approximately 1 in 100 cycles where a known failure mechanism is documented in the literature or datasheet and the design only partially mitigates it. A high occurrence score (7 to 8) indicates approximately 1 in 10 cycles where the failure mechanism is dominant and the current design does not adequately address it, and a very high score (9 to 10) indicates greater than 1 in 2 cycles where failure is expected under normal use without corrective design action. It is important to note that several occurrence scores in this FMEA, particularly for FM-01 (FSR adhesive delamination) and FM-03 (moisture ingress), are based on engineering judgment rather than empirical test data. These scores are conservative estimates that will be replaced with test-derived values following bench testing under simulated ambulatory and moisture exposure conditions during Design Verification. Table 13 presents the full occurrence scale with the estimation basis for each range.

Table 13: Occurrence Scale and NeuroFoot Basis

Score	Frequency Estimate	Basis
1 to 2	Remote: fewer than 1 in 10,000 operating cycles	Component-level specification provides very high reliability assurance
3 to 4	Low: approximately 1 in 1,000 cycles	Component meets specification; failure possible but infrequent under normal use
5 to 6	Moderate: approximately 1 in 100 cycles	Known failure mechanism documented in literature or datasheet; design partially mitigates
7 to 8	High: approximately 1 in 10 cycles	Failure mechanism is dominant and current design does not adequately address it
9 to 10	Very High: greater than 1 in 2 cycles	Failure is expected under normal use without corrective design action

Detection reflects the ability of existing design controls to identify the failure mode before it reaches the patient without correction. A higher detection score indicates a lower probability of detection, and a score of 10 represents a complete inability to detect the failure. At the low end of the scale (score 1 to 2), detection is almost certain because the device firmware or hardware automatically detects the failure and triggers a visible or audible alert before patient impact. A score of 3 to 4 indicates high detectability, where the patient or user would likely notice anomalous device behavior within one use session. Moderate detectability (score 5 to 6) means that detection requires active inspection, a calibration check, or app review and is not self-evident during routine use. Low detectability (score 7 to 8) applies to failures that are gradual or latent, where the patient is unlikely to notice degraded performance without dedicated monitoring. Remote detectability (score 9 to 10) describes silent failures where the device gives no indication of malfunction and the patient cannot distinguish an operational device from a non-operational one. This last category is of particular concern for the NeuroFoot population, because patients with loss of protective sensation have no sensory fallback to independently verify that monitoring is active. Table 14 presents the full detection scale with the detection mechanism for each range.

Table 14: Detection Scale and NeuroFoot Detection Mechanism

Score	Detectability	Mechanism
1 to 2	Almost certain	Automatic detection by device firmware or hardware; failure triggers visible or audible alert before patient impact

Score	Detectability	Mechanism
3 to 4	High	Patient or user would likely notice anomalous device behavior within one use session
5 to 6	Moderate	Detection requires active inspection, calibration check, or app review; not self-evident during routine use
7 to 8	Low	Failure is gradual or latent; patient is unlikely to notice degraded performance without dedicated monitoring
9 to 10	Remote	Silent failure; device gives no indication of malfunction; patient cannot distinguish operational from non-operational states

2.2.b. Risk Acceptance Criteria and FMEA Results

The risk acceptance thresholds applied to prioritize corrective action are established consistent with ISO 14971:2019 Section 7 (Risk Evaluation) and are applied to initial RPN values prior to implementation of recommended actions. An RPN of 200 to 1,000 is classified as High Risk, meaning corrective action is required before Design Verification and the revised RPN must fall below 200 following implementation of recommended controls. An RPN of 100 to 199 is classified as Medium Risk, meaning risk reduction is recommended and recommended actions are documented and addressed in the Design Verification plan. An RPN of 1 to 99 is classified as Low Risk, meaning the risk is acceptable with existing controls and will be monitored and reassessed if design changes are made. These thresholds are intentionally conservative at the PDR stage: the High Risk threshold of 200 ensures that the two most consequential failure modes (FM-01 and FM-03, discussed below) receive mandatory corrective action rather than being deferred. Table 15 summarizes the risk acceptance criteria.

Table 15: Risk Acceptance Criteria of Neurofoot

RPN Range	Risk Level	Required Action
200 to 1000	High	Corrective action required before Design Verification. Revised RPN must fall below 200 following implementation of recommended controls.
100 to 199	Medium	Risk reduction recommended. Recommended actions documented and addressed in the Design Verification plan.
1 to 99	Low	Risk acceptable with existing controls. Monitor and reassess if design changes are made.

The FMEA identifies six failure modes spanning sensor integrity, electromagnetic interference, moisture protection, wireless communication, power management, and environmental sensing.

Of these six, two carry initial RPN scores above 200 and are classified as High Risk. FM-01 (FSR adhesive delamination, RPN 315) receives the highest score because it combines a severity of 9 with moderate occurrence and low detectability. If the adhesive bond between an FSR and the PCB substrate fails, the affected sensor underreports planar loading, and the patient receives no alert during sustained overload. The patient cannot verify sensor accuracy independently, and the failure is entirely silent. FM-03 (moisture ingress and short circuit, RPN 245) is the second High Risk failure mode. Incomplete conformal coating coverage or inadequate foam resealing during assembly allows sweat or external moisture to reach the PCB, causing intermittent or complete device failure. The patient may be unaware that monitoring has stopped.

Three failure modes are classified as Medium Risk. FM-02 (60 Hz EMI false alerts, RPN 120) arises when environmental electromagnetic interference causes spurious threshold crossings. The clinical concern is not the false alert itself but the behavioral consequence: repeated false alerts cause alert fatigue, and a patient who begins ignoring warnings is at elevated risk of missing a genuine sustained-loading event. FM-04 (BLE disconnection during ambulation, RPN 168) interrupts caregiver app notifications during walking, though the onboard buzzer and vibration motor continue to operate independently of BLE. FM-05 (battery depletion without user notification, RPN 175) results in complete monitoring loss when the patient forgets to charge the device and the low-battery warning is missed, a scenario that is particularly relevant for patients with cognitive comorbidities or limited smartphone familiarity.

One failure mode is classified as Low Risk. FM-06 (HS300x cold boot initialization failure, RPN 24) occurs when the environmental sensor fails to initialize on the first firmware boot attempt. The firmware retry logic resolves the initialization before the device is ready for use, and the primary pressure monitoring and alerting functions are entirely unaffected. The clinician is notified of sensor health status through the companion app. Table 16 presents the full FMEA with S, O, D scores, RPN values, risk levels, and clinical risk justifications for all six failure modes. Red shading indicates an RPN of 200 or above (High Risk), yellow shading indicates an RPN of 100 to 199 (Medium Risk), and green shading indicates an RPN below 100 (Low Risk).

Table 16: Failure Mode and Effects Analysis Table of Neurofoot

FM ID	Failure Mode	S	O	D	RPN	Risk Level	Clinical Risk Justification
FM-01	FSR Adhesive Delamination	9	5	7	**315**	**HIGH**	Patient cannot verify accuracy; silent overload missed; loss of protective warning
FM-02	60 Hz EMI False Alerts	5	4	6	**120**	**MEDIUM**	Alert fatigue leads to patient ignoring legitimate warnings; behavioral desensitization

FM ID	Failure Mode	S	O	D	RPN	Risk Level	Clinical Risk Justification
FM-03	Moisture Ingress / Short Circuit	7	5	7	**245**	**HIGH**	Intermittent or complete device failure; silent monitoring loss during wear
FM-04	BLE Disconnection (ambulatory)	7	4	6	**168**	**MEDIUM**	No alerts delivered during potentially dangerous loading; LED indicator not visible
FM-05	Battery Depletion / User Failure	7	5	5	**175**	**MEDIUM**	Complete monitoring loss unknown to patient; extended unmonitored wear
FM-06	HS300x Cold Boot Init Failure	2	4	3	**24**	**LOW**	Incorrect environmental data on first read; firmware retry loop resolves; no alert impact

Each failure mode carries both a manufacturing risk and a user risk, and these two dimensions are not always equal in severity. FM-01, for example, is High Risk on both dimensions: insufficient adhesive application during assembly causes premature delamination (manufacturing), and a delaminated sensor silently underreports plantar loading during use (user). FM-04, by contrast, is Low Risk from a manufacturing perspective because BLE antenna placement and sandal layer shielding are design considerations rather than assembly defects, but Medium Risk from a user perspective because BLE dropout during walking interrupts caregiver notifications. FM-05 is similarly asymmetric: the manufacturing risk is limited to incorrect battery connector polarity during assembly, which is controlled by polarity marking and pre-embed testing, but the user risk is High because a patient who forgets to charge the device faces complete unmonitored wear with no sensory fallback. Understanding this asymmetry is important for directing risk controls to the correct phase of the device lifecycle. Manufacturing-dominant risks are addressed through assembly procedure controls, inspection gates, and process documentation in Section 1.2.d. User-dominant risks are addressed through firmware design controls, companion app notifications, and patient/caregiver education materials. Table 17 presents the full risk classification by type, distinguishing manufacturing risk from user risk for each failure mode.

Table 17: Risk Classification by Type (Manufacturing Risk vs. User Risk)

FM ID	Failure Mode	Manufacturing Risk	User Risk
FM-01	FSR adhesive delamination	HIGH. Insufficient adhesive application, incorrect adhesive type, or inadequate cure time during assembly causes premature delamination. Controlled by assembly procedure and post-cure inspection gate.	HIGH. Delaminated sensor underreports plantar loading during use; patient receives no alert during sustained overload. Silent failure with no sensory fallback.
FM-02	60 Hz EMI false alerts	MEDIUM. Missing or incorrect pull-down resistor values during PCB population increase EMI susceptibility. Controlled by BOM specification and component verification.	MEDIUM. False threshold alerts during use cause alert fatigue; patients may begin ignoring warnings, increasing risk of missed genuine alert.
FM-03	Moisture ingress / short circuit	HIGH. Incomplete conformal coating coverage or inadequate foam resealing during assembly allows moisture ingress. Controlled by mandatory coating inspection and waterproof adhesive specification.	HIGH. Moisture ingress during daily use (sweat, rain) causes intermittent or complete device failure. Patients may be unaware monitoring has stopped.
FM-04	BLE disconnection during ambulation	LOW. Not a manufacturing risk. BLE antenna placement and sandal layer shielding are design considerations; firmware handles reconnection automatically.	MEDIUM. BLE dropout during walking interrupts caregiver app notifications. Mitigated by an onboard buzzer/vibration alert which operates independently of BLE.
FM-05	Battery depletion without user notification	LOW. Battery capacity and connector type are design/BOM decisions. Manufacturing risk is limited to incorrect battery connector polarity during assembly, controlled by polarity marking and pre-embed test.	HIGH. Patient forgets to charge; device silently depletes mid-use with no monitoring. Patients with cognitive comorbidities or limited smartphone familiarity are at elevated risk of missing low-battery warnings.
FM-06	HS300x cold boot initialization failure	LOW. Firmware initialization sequence order is a software design issue, not a manufacturing defect. Detected and logged during Stage 3 pre-embed electrical verification.	LOW. Firmware retry logic resolves initialization before the device is ready for use. Primary pressure monitoring and alerting functions are unaffected. Clinician notified via app sensor health status.

ISO 14971:2019 establishes the requirements for a risk management process applied throughout the device lifecycle. The revised FMEA, comprising six failure modes, is evaluated below against each applicable clause of the standard to identify current compliance status and remaining gaps that must be closed before Design Verification. The purpose of this assessment is not to claim full compliance at the PDR stage, but to establish a clear record of what has been completed, what is partially addressed, and what remains open. Several clauses are partially addressed through the FMEA itself, while others require formal documentation that does not yet exist, such as a standalone risk management plan (Clause 4), a risk-benefit analysis (Clause 10), and a risk management report (Clause 11). Table 18 summarizes the clause-by-clause compliance status and gap actions.

Table 18: ISO 14971:2019 Clause, current status and action for Neurofoot

ISO 14971:2019 Clause	Requirement Summary	Current Status	Gap or Action
Clause 4: Risk Management Plan	Establish a documented plan defining scope, risk acceptability criteria, and responsibilities	Partially addressed. Scope covers six failure modes across sensor integrity, EMI, moisture protection, wireless communication, power management, and environmental sensing. Scoring criteria defined in Section 2.2a; responsibilities are implicitly assigned but not formally documented	Formalize a risk management plan document that names responsible individuals and defines review milestones aligned with the ISO 13485 design review schedule
Clause 5: Hazard Identification	Systematically identify all foreseeable hazards associated with the device and its intended use	Six failure modes identified. FM-01 (FSR adhesive delamination) and FM-03 (moisture ingress) address physical and environmental hazards. FM-02 (60 Hz EMI) and FM-04 (BLE disconnection) address electromagnetic and communication hazards. FM-05 (battery depletion) and FM-06 (HS300x cold boot failure) address power and sensor initialization hazards	Expand hazard identification to include foreseeable misuse scenarios (e.g., patient continues ambulation during BLE dropout) and use environment hazards beyond moisture, consistent with ISO 14971 Annex C guidance
Clause 6: Risk Estimation	Estimate probability and severity of harm for each hazard	Quantitative scoring using S, O, D scale with defined criteria. FM-01 (RPN 315) and FM-03 (RPN 245) carry the highest initial risk scores. FM-06 (RPN 24) is rated low, with	Occurrence scores for FM-01 and FM-03 are based on engineering judgment at PDR stage. Bench testing under simulated ambulatory conditions and moisture exposure

		firmware retry logic limiting clinical impact	should replace judgment-based scores at Design Verification
Clause 7: Risk Evaluation	Determine whether each risk is acceptable based on predefined criteria	Risk acceptance thresholds defined: RPN 200 or above = High Risk requiring action. FM-01 (315) and FM-03 (245) exceed this threshold. FM-02, FM-04, and FM-05 are Medium Risk. FM-06 is Low Risk and acceptable with existing controls	Risk acceptance criteria should be formally approved by a responsible individual and recorded in the risk management plan rather than in the FMEA table alone
Clause 8: Risk Control	Implement and verify risk control measures in order of priority: inherent safety by design, protective measures, information for safety	Recommended actions are documented for all failure modes. Inherent design controls include moisture-resistant enclosure design (FM-03) and EMI filtering (FM-02). Informational controls include low-battery notification (FM-05) and firmware watchdog reset (FM-06)	Confirm that risk control measures are implemented before Design Verification. Verify effectiveness through environmental and electrical testing and update revised RPN values with empirical test data
Clause 9: Residual Risk Evaluation	Evaluate whether residual risk after controls is acceptable	Revised RPN values are estimated for all failure modes following recommended actions. Residual estimates for FM-01 and FM-03 are projected to fall below the high-risk threshold of 200 following enclosure and adhesive redesign	Residual risk estimates are projections only. Actual revised RPN values must be recalculated using post-control test data. A formal residual risk acceptance statement should be recorded in the risk management report
Clause 10: Risk-Benefit Analysis	Justify acceptance of residual risk by demonstrating clinical benefit exceeds risk	Not yet formally documented at PDR stage	Prepare a risk-benefit analysis referencing Abbott et al. (2019, Lancet Digital Health) reporting 71% reduction in DFU recurrence with analogous plantar pressure alerting technology. This analysis should be completed before Design Validation
Clause 11: Risk Management Report	Summarize the risk management process outcomes	Not yet prepared; FMEA is the primary risk documentation at this stage	Produce a formal risk management report at the conclusion of Design Verification, consolidating the FMEA, residual risk evaluations, and risk-benefit analysis

The following FDA guidance documents are directly relevant to the risk analysis for NeuroFoot. Each document imposes requirements or recommendations that intersect with one or more of the six failure modes identified in the FMEA. The evaluation below maps each guidance document to the specific failure modes it addresses, identifies the current compliance status, and defines the gap actions required before advancing to Design Verification or premarket submission. 21 CFR 820.30(g) requires that the device be validated under simulated or actual conditions of use, and the high-risk failure modes FM-01 and FM-03 directly inform the minimum scope of that validation testing. The FDA's 2016 Human Factors guidance is particularly relevant to FM-05, because battery depletion without user notification is a use-related silent failure that can only be characterized through formative usability evaluation with representative users from the IWGDF Risk 2 to 3 population. The 2013 Design Considerations guidance links risk identification to the scope of clinical or non-clinical performance testing, reinforcing the need for bench validation of adhesive integrity and enclosure sealing. The 2023 Cybersecurity guidance applies to FM-04 and to the BLE communication pathway more broadly, requiring threat modeling, a software bill of materials, and a vulnerability disclosure plan. Table 19 presents the full FDA guidance evaluation.

Table 19: Potential FDA Guidance or Regulation for Neurofoot Device

FDA Guidance or Regulation	Requirement Summary	Current Status	Gap or Action
21 CFR 820.30(g): Design Validation	Device must be validated under simulated or actual conditions of use to ensure it meets user needs and intended uses	Design Validation plan not yet established at PDR stage. Risk analysis informs the scope of required validation testing	Use high-risk failure modes FM-01 (FSR adhesive delamination, RPN 315) and FM-03 (moisture ingress, RPN 245) to define minimum validation test cases. Validation should include mechanical cycling under ambulatory loads and moisture exposure at physiologically relevant humidity levels
FDA Guidance: Applying Human Factors and Usability Engineering (2016)	Human factors engineering must address use-related risks for devices used in home settings by non-clinical users	FM-05 (battery depletion with no user notification) identifies a use-related silent failure mode. No formative human factors study has been conducted yet	Conduct formative usability evaluation with representative users from the IWGDF Risk 2 to 3 population, focusing on alert recognition and battery status interpretation. Update FM-05 occurrence and detection scores based on observed use errors

FDA Guidance: Design Considerations for Pivotal Clinical Investigations (2013)	Risk identification should inform the scope of clinical or non-clinical performance testing	FM-01 (FSR delamination) and FM-03 (moisture ingress) are high-risk failure modes requiring bench validation of enclosure and adhesive performance under real-world mechanical and environmental conditions	Define bench testing protocols for FSR adhesive integrity under ambulatory load cycling and for enclosure moisture resistance. Reference applicable standards (e.g., IEC 60529 for ingress protection rating) in the Design Verification plan
FDA 2023 Cybersecurity Guidance: Cybersecurity in Medical Devices	Connected devices must address cybersecurity risks in premarket submissions, including threat modeling, SBOM, and vulnerability disclosure	FM-04 (BLE disconnection during ambulation) is identified as a medium-risk failure mode affecting alert delivery. BLE 5.0 is implemented, but reconnection behavior and alert fallback during dropout are open design items	Define and implement a BLE reconnection protocol with local fallback alerting. Confirm AES-128 encryption implementation status. Prepare SBOM and threat model (STRIDE analysis) covering BLE and smartphone app pathways. Document cybersecurity architecture in the De Novo or 510(k) submission package if a premarket pathway requiring submission is pursued

Based on the FMEA results and the ISO 14971 and FDA alignment assessment, the following actions are recommended in order of priority before advancing to Design Verification.

The highest-priority corrective action is to redesign the FSR adhesive interface and validate mechanical integrity (FM-01, RPN 315). FSR adhesive delamination under ambulatory loading represents a silent sensor failure that cannot be detected by the patient. A cycling test protocol should be designed to characterize adhesive bond strength and FSR output stability over a minimum of 5,000 loading cycles representative of daily ambulatory use. Results should be used to qualify the adhesive specification and update occurrence and detection scores.

The second priority is to validate moisture ingress protection and enclosure sealing (FM-03, RPN 245). Moisture ingress is the second highest-risk failure mode and can result in intermittent or complete device failure during wear. An enclosure sealing specification appropriate for the footwear environment should be defined, and ingress protection testing should be conducted per IEC 60529 or an equivalent standard. The test protocol should replicate physiologically relevant sweat exposure and repeated mechanical deformation of the insole enclosure.

The third priority is to implement a local fallback alert for BLE disconnection during ambulation (FM-04, RPN 168). BLE dropouts during walking are a documented risk in the 2.4 GHz ISM band. Because disconnection may be invisible to the patient, the current architecture has a single point of failure in alert delivery. A hardware alert mechanism within the insole enclosure should be designed to activate when BLE connection is not confirmed within a defined timeout window. This eliminates dependence on smartphone connectivity for real-time safety alerting.

The fourth priority is to implement battery status monitoring and low-power alerting (FM-05, RPN 175). Battery depletion without user notification results in complete monitoring loss that is unknown to the patient, a clinical risk that is not currently covered by any existing design control. The nRF52840 battery voltage monitoring peripheral should be configured, and firmware thresholds for low-battery push notifications and LED indication should be defined at the 20% and 10% levels.

The fifth priority is to add EMI filtering for 60 Hz interference (FM-02, RPN 120). Sixty hertz interference from power line sources can generate false pressure alerts, contributing to alert fatigue and behavioral desensitization over time. Hardware low-pass filtering on FSR signal lines should be implemented and validated for suppression of 60 Hz noise below the alert threshold during bench testing under representative electrical environments.

The sixth and lowest priority is to confirm HS300x cold boot initialization behavior in firmware qualification testing (FM-06, RPN 24). Although FM-06 is rated low risk due to firmware retry logic that resolves the initialization failure before any alert impact, the retry behavior must be explicitly verified in the software qualification test plan. The expected initialization sequence, retry count, and failure mode should be documented in the IEC 62304 software requirements specification.

2.3. Mitigation & Design Controls

2.3.a. Risk Control Measures

The following risk control measures address each failure mode identified in Section 2.2. Controls are organized by failure mode and categorized as hardware, firmware, or procedural interventions. All controls are designed with the diabetic peripheral neuropathy patient population in mind, specifically users who cannot self-report device-induced discomfort and who have no sensory fallback if the device fails silently.

The highest-priority control addresses FM-01 (FSR adhesive delamination, RPN 315). The original design relied solely on solder connections for FSR retention, which created a single point of mechanical failure under repeated compressive loading. The primary hardware change is to bond each FSR sensor to its PCB pad using epoxy adhesive in addition to solder, so that mechanical retention is independent of solder joint integrity. Conformal coat is extended over sensor leads and the surrounding PCB area to resist moisture-driven bond degradation, which is a secondary failure pathway that weakens the adhesive interface over time. On the procedural side, a sensor adhesion endurance test is included in final QC: repeated compressive loading is applied to confirm that sensors remain bonded and reading correctly after simulated ambulatory cycling. A 30-day recalibration and inspection schedule is defined that includes physical inspection of sensor bonding through foam surface feel, since any delamination large enough to affect readings will produce a detectable change in surface compliance.

The second control set addresses FM-03 (moisture ingress and short circuit, RPN 245). The original prototype had no explicit waterproofing strategy. The primary hardware change is to apply a full conformal coat to all PCB traces, solder joints, and component bodies before foam embedding, which serves as the primary moisture barrier. The foam sandal is resealed using a waterproof adhesive specified in the bill of materials, forming a secondary moisture barrier at the cavity interface. Foam absorbs sweat over time, so the adhesive seal must be continuous and free of gaps. During validation, an assembled device is exposed to saline solution simulation to confirm no electrical failure or sensor drift. End-of-life criteria are defined based on moisture exposure history and foam degradation inspection, because the moisture barriers degrade with use and cannot be refreshed without disassembly.

The third control set addresses FM-02 (60 Hz EMI false alerts, RPN 120). The original design had no explicit analog trace routing rules or firmware filtering. On the hardware side, all analog signal traces (FSR outputs, multiplexer common I/O) are routed as short as possible and separated from digital switching lines and power traces on the PCB. Decoupling capacitors are added at multiplexer supply pins and at the Arduino analog reference pin to suppress high-frequency noise. On the firmware side, a rolling median filter is applied across all 16 channels to reject transient spikes that do not represent sustained plantar loading. Alert debounce logic requires the threshold to be exceeded continuously for a minimum number of consecutive samples before triggering an alert, eliminating single-sample false positives from environmental interference.

The fourth control set addresses FM-04 (BLE disconnection during ambulation, RPN 168). The fundamental architectural decision is that the onboard buzzer and vibration motor operate entirely independently of BLE connection status, so that ulcer risk alerts function without a connected phone under all circumstances. This decouples the safety-critical alert pathway from the wireless link. BLE reconnection attempts run automatically in the background without interrupting sensor polling or alert logic. The companion app displays connection status prominently and alerts the caregiver if the device goes offline unexpectedly for more than a configurable duration, providing a secondary notification pathway for extended disconnections.

The fifth control set addresses FM-05 (battery depletion without user notification, RPN 175). Two firmware-level battery alerts are implemented: a low-battery alert at 20% remaining charge using a distinct alert pattern that the patient can differentiate from an ulcer warning, and a critical battery alert at 5% remaining that also triggers a caregiver app notification. The hardware charging interface uses a magnetic or USB-C connector that is easy to locate and connect without looking, accommodating reduced dexterity common in diabetic patients. The caregiver app displays real-time battery level and sends a push notification when charging is needed, providing a secondary reminder pathway for patients who may not notice or understand the onboard alert.

The sixth control set addresses FM-06 (HS300x cold boot initialization failure, RPN 24). The firmware implements an automatic retry loop with a maximum of three attempts on HS300x initialization at boot. If all retries fail, the failure is logged to onboard memory and an error code is transmitted to the companion app. The HS300x is classified as a non-critical sensor in the system architecture, meaning the device continues operating for primary pressure monitoring and alerting functions even if the temperature and humidity sensor fails to initialize. The companion app displays sensor health status so that the caregiver or clinician is aware of the degraded sensor state and can schedule maintenance if needed.

Beyond the six FMEA failure modes, three additional risk categories are addressed through design controls. The first is silent device failure, which represents the most significant architectural design change in the system. A hardware watchdog timer is implemented so that if the main firmware loop does not reset the watchdog within a defined interval, the microcontroller automatically resets rather than hanging in an unresponsive state. A heartbeat signal is transmitted via BLE to the companion app every 60 seconds, and the absence of a heartbeat triggers a caregiver alert. An onboard status LED provides visual indication of device operational state, with distinct patterns for normal operation, alert active, low battery, and fault condition. At startup, a self-test routine runs across all 16 sensor channels, the IMU, and all alert outputs. If critical components are non-responsive, the device fails to operate rather than running silently in a degraded state. These changes collectively convert what would otherwise be a silent failure into an observable, alertable event.

The second additional category is alert fatigue, which is a behavioral risk rather than a hardware failure but carries real clinical consequences if patients begin ignoring legitimate warnings. The alert threshold is set based on clinically referenced sustained pressure duration values from the diabetic foot ulcer literature rather than arbitrary engineering defaults. Pressure accumulation is gated to the stance phase only using IMU gait phase detection, which eliminates false accumulation during swing phase, sitting, and non-weight-bearing periods. The rolling median filter applied across all channels (described under FM-02) also serves to reject sensor noise and drift-induced false readings before threshold comparison. Distinct alert patterns are defined for ulcer risk, low battery, and device fault, ensuring that each alert type carries unambiguous meaning so that patients do not become desensitized to a single repeating stimulus.

The third additional category is PCB shift inside the foam enclosure. The assembly procedure requires structural epoxy applied across the full PCB contact surface rather than spot applications, which distributes bonding force evenly and prevents rocking or rotation. After cure, lateral force is applied to the PCB to confirm no movement before proceeding to foam resealing. A PCB positional check is included in the 30-day recalibration inspection: sensor zone alignment is verified by comparing readings under standardized loading to the baseline recorded during initial assembly.

A closely related concern is foam compression set, which refers to the gradual permanent deformation of the EVA foam under repeated loading. This is addressed at the material specification level by requiring documented compression set resistance rating in the bill of materials, selecting a foam grade with low compression set characteristics under repeated loading. A recalibration schedule is defined requiring cross-sensor normalization recalibration every 30 days or whenever a user reports changes in device feel. The firmware implements a baseline drift detection algorithm that prompts the user or caregiver to schedule recalibration if resting sensor readings shift beyond a defined threshold relative to original calibration values. Device end-of-life is defined based on a measurable foam compression set: the device is retired when foam thickness in high-load zones has compressed beyond the acceptable tolerance established during design verification.

2.3.b. Design Changes and Risk Assessment Procedures

This section describes the specific design changes made in response to identified failure modes and defines the procedure by which current risk levels are assessed and monitored throughout the device lifecycle.

The design changes are summarized by failure mode to provide a concise record of what changed and why. For FM-01 (FSR adhesive delamination), the original design relied solely on solder connections for FSR retention, and the change was to add epoxy bonding as a mechanical retention layer independent of solder joint integrity, with conformal coat extended over sensor leads and surrounding pad areas. For FM-02 (60 Hz EMI false alerts), the original design had no explicit analog trace routing rules or firmware filtering, and the changes were to add PCB layout rules requiring short, separated analog traces with decoupling capacitors, and to update the firmware with a rolling median filter and alert debounce logic. For FM-03 (moisture ingress), the original prototype had no explicit waterproofing, and the changes were to add mandatory conformal coating before embedding as a process step with a pass/fail gate and to specify waterproof foam resealing adhesive in the bill of materials. For FM-04 (BLE disconnection), the alert architecture was decoupled from BLE so that the onboard buzzer and vibration motor operate independently; no change was made to BLE hardware, and the change is architectural in how firmware delivers alerts. For FM-05 (battery depletion), a two-level battery alert system at 20% and 5% was added in firmware, a magnetic charging connector was specified to reduce user dexterity burden, and caregiver app battery monitoring was added. For FM-06 (HS300x cold boot failure), the firmware was updated with a three-attempt retry loop, and the HS300x was reclassified as a non-critical sensor so that primary functions continue independently of its state.

The most significant architectural design change addressed silent device failure as a category. A watchdog timer, heartbeat BLE signal, startup self-test routine, and status LED were all added. Together, these changes convert what would be a silent, undetectable failure into an observable event that triggers an alert to the patient, caregiver, or both.

For alert fatigue, the alert threshold was updated to reference clinical literature on diabetic foot ulcer pressure duration, IMU gait phase gating was added to prevent false accumulation during non-weight-bearing periods, and distinct alert patterns were defined for each alert type. For PCB shift inside foam, the assembly procedure was updated to require structural epoxy across the full PCB contact surface, and a post-cure lateral force check was added as a mandatory inspection step. For foam compression set, the foam specification was updated to require documented compression set resistance rating, a recalibration schedule was added to device documentation, a baseline drift detection algorithm was added to firmware, and end-of-life criteria were defined based on measurable foam compression.

Current risk is assessed through a combination of design-phase analysis, assembly-phase testing, and in-service monitoring that together provide continuous coverage across the device lifecycle. During the design phase, Risk Priority Number analysis is performed for each identified failure mode using the Severity, Occurrence, and Detection ratings defined in Section 2.2.a. RPN is calculated before and after control measures are applied to quantify risk reduction and to confirm that all High Risk failure modes fall below the 200 threshold after controls are implemented.

During the assembly phase, each assembly stage includes defined pass/fail inspection gates as described in the Section 1.2.d assembly flowchart. Failure at any gate initiates a documented non-conformance record that is reviewed before the unit proceeds or is scrapped. Pre-embedding electrical test results and final QC test results are logged per unit serial number, creating a traceable quality record for every assembled device.

During in-service use, the companion app logs device health data including alert frequency, sensor baseline drift, BLE connectivity history, battery charge cycles, and fault events. Anomalies trigger caregiver notification and are reviewed at the scheduled 30-day recalibration appointment. At each recalibration inspection, the device undergoes physical inspection for surface smoothness and PCB positional stability, sensor baseline recalibration, cross-sensor normalization, and foam compression assessment. Findings are logged and compared to prior inspection records to track drift trends across the device lifetime.

The device is retired when any of the following conditions are met: foam compression in high-load zones exceeds the tolerance defined during design verification, sensor baseline drift exceeds the correction range of the recalibration procedure, recalibration fails to restore readings within specification, or physical inspection reveals delamination, moisture damage, or surface irregularity. These end-of-life criteria ensure that a degraded device is removed from service before its monitoring accuracy falls below the clinically acceptable threshold.

2.3.c. Risk Reduction Quantification

Risk reduction is quantified by comparing Risk Priority Number values before and after control measures are applied to each failure mode. Each failure mode is rated on three dimensions: Severity (S), which captures the impact on patient safety if the failure occurs; Probability (P), which captures the likelihood of occurrence under normal use conditions; and Detectability (D), which captures the ease of detecting the failure before it causes patient harm. RPN is calculated as the product of all three scores.

Severity is rated on a scale of 1 to 5. A score of 1 indicates a negligible failure with no patient impact. A score of 2 indicates a minor failure causing device inconvenience but no clinical consequence. A score of 3 indicates a moderate failure that degrades monitoring capability but carries low clinical risk. A score of 4 indicates a serious failure resulting in a missed ulcer warning or a false alarm with clinical consequence. A score of 5 indicates a critical failure causing direct patient harm or complete loss of the device's protective function.

Probability is also rated on a scale of 1 to 5. A score of 1 indicates a remote failure that is unlikely under normal conditions. A score of 2 indicates a low-probability failure that is possible under edge conditions. A score of 3 indicates a moderate probability, meaning the failure is expected occasionally over the device lifetime. A score of 4 indicates a high probability, meaning the failure is expected regularly under normal use. A score of 5 indicates a near-certain failure that is expected frequently or inevitably without corrective design action.

Detectability is rated inversely, so that a higher score indicates a harder-to-detect failure. A score of 1 indicates an obvious failure that is immediately apparent to the patient, caregiver, or device. A score of 2 indicates a failure that will probably be noticed within a short period. A score of 3 indicates a failure that may be noticed if the user is attentive. A score of 4 indicates a failure that is unlikely to be noticed without dedicated inspection. A score of 5 indicates a silent failure with no observable indicator, which is the most dangerous category for the NeuroFoot population because patients with loss of protective sensation have no sensory fallback to detect degraded device performance.

The target post-control RPN is 20 or below for all failure modes. Any failure mode with a post-control RPN above 20 requires additional design review before the device can advance to Design Verification.

The pre-control and post-control RPN values for all ten failure modes are presented in Table 20. Primed values (S', P', D', RPN') reflect the ratings after control measures described in Section 2.3.a have been applied.

Table 20: Risk Reduction Summary (Pre-Control and Post-Control RPN)

Failure Mode	S	P	D	RPN	Control Measure	S'	P'	D'	RPN'
FSR Adhesive Delamination	4	3	3	36	Epoxy bond + conformal coat; endurance test at assembly	4	2	2	16
60 Hz EMI False Alerts	3	3	2	18	Shielded analog traces; firmware spike filter; debounce threshold	3	2	1	6
Moisture Ingress / Short Circuit	5	4	2	40	Conformal coat pre-embedding; foam resealed with waterproof adhesive	5	2	2	20
BLE Disconnection (ambulatory)	2	4	1	8	Onboard buzzer/vibration operates independently of BLE	2	4	1	8
Battery Depletion / User Failure	4	4	2	32	Low battery alert at 20%; caregiver app notification; magnetic charge port	4	2	1	8
HS300x Cold Boot Init Failure	2	2	2	8	Firmware retry loop (3x); non-critical sensor; log failure to app	1	2	1	2
Silent Device Failure	5	3	4	60	Watchdog timer resets firmware; heartbeat LED/app ping every 60s	5	2	2	20
Alert Fatigue	4	3	3	36	Clinically-referenced threshold; rolling median filter; alert debounce	4	2	2	16
PCB Shift Inside Foam	4	2	4	32	Epoxy bonding of PCB to cavity; shift inspection at final QC	4	1	2	8
Foam Compression Set	3	4	3	36	Recalibration schedule every 30 days; high-durometer foam spec	3	2	2	12

The two highest pre-control RPNs belong to silent device failure (RPN 60) and moisture ingress (RPN 40). Silent device failure received the highest initial score because it combines a critical severity of 5 with a detectability of 4, reflecting the fact that without a watchdog timer or heartbeat signal, a frozen or crashed device gives no outward indication of failure. The control measures for this failure mode, including the watchdog timer, BLE heartbeat, startup self-test, and status LED, reduce both probability (from 3 to 2) and detectability (from 4 to 2), bringing the post-control RPN to 20. Moisture ingress carries a critical severity of 5 that cannot be reduced through controls because moisture reaching the PCB will cause electrical failure regardless of design mitigations. The controls instead target probability (conformal coating and waterproof adhesive reduce occurrence from 4 to 2), bringing the post-control RPN to 20. Both of these failure modes sit at the upper boundary of the acceptable range, which reflects the inherent seriousness of their consequences rather than inadequate control.

FSR adhesive delamination and alert fatigue both start at an RPN of 36 and are reduced to 16 after controls. For delamination, the addition of epoxy bonding independent of solder joint integrity reduces probability from 3 to 2, and the inclusion of an endurance test during assembly reduces detectability from 3 to 2, while severity remains at 4 because a delaminated sensor still silently underreports pressure. For alert fatigue, clinically referenced thresholds and IMU gait phase gating reduce the probability of false alerts from 3 to 2, and the rolling median filter with debounce logic reduces detectability from 3 to 2 by catching noise-induced false readings before they reach the alert logic.

Battery depletion starts at an RPN of 32 and drops to 8 after controls. The two-level alert system at 20% and 5% charge, combined with caregiver app notification, reduces probability from 4 to 2 (because patients are now warned before depletion occurs) and detectability from 2 to 1 (because the alerts are impossible to miss). PCB shift inside foam also starts at 32 and drops to 8. Full-surface epoxy bonding and the post-cure lateral force check reduce probability from 2 to 1, and the 30-day recalibration positional check reduces detectability from 4 to 2.

Foam compression set starts at an RPN of 36 and is reduced to 12. Specifying a high-durometer foam with documented compression set resistance and implementing the 30-day recalibration schedule with baseline drift detection reduces probability from 4 to 2 and detectability from 3 to 2. Severity remains at 3 because gradual foam compression degrades monitoring accuracy but does not produce a complete loss of function.

The 60 Hz EMI false alerts mode starts at 18 and drops to 6 after shielded analog trace routing, decoupling capacitors, and firmware filtering are applied. Probability drops from 3 to 2, and detectability drops from 2 to 1 because the firmware filter now catches and suppresses interference before it can trigger a false alert.

BLE disconnection during ambulation is the only failure mode whose RPN does not change after controls are applied, remaining at 8 both before and after. This is because the primary control, which is independent onboard alerting via buzzer and vibration motor, eliminates the clinical consequence of a BLE dropout entirely rather than reducing its probability or detectability. The BLE link will still drop occasionally in the 2.4 GHz ISM band (probability remains at 4), and the dropout itself is still not immediately obvious to the patient (detectability remains at 1), but neither of these matters for patient safety because the alert pathway no longer depends on BLE connectivity. Severity is the binding constraint at a score of 2, and it is accepted given the architectural mitigation.

The HS300x cold boot initialization failure carries the lowest pre-control RPN at 8 and drops to 2 after controls. The firmware retry loop resolves the initialization in nearly all cases, and reclassifying the HS300x as a non-critical sensor means that even a persistent initialization failure has negligible clinical impact, reducing severity from 2 to 1.

All ten failure modes achieve a post-control RPN of 20 or below, meeting the target threshold established for Design Verification. No failure mode requires additional design review at this stage.

3. Work Cited

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4. Appendix A: Predicate Device FDA Clearance Summary

SmartStep System (K060150)

Field	Detail
510(k) Number	K060150
Clearance Date	February 23, 2006
Applicant	Andante Medical Devices Ltd., Yoqneam Illit, Israel
Predicate Device	K023161 (earlier SmartStep model)
Regulation Number	21 CFR 890.5575
Product Code	IRN (Powered External Limb Overload Warning Device)
Classification	Class II
Review Panel	Physical Medicine
Standards Referenced	IEC 60601-1, IEC 60601-1-2, AAMI/ISO 14971-1

Indications for Use (Verbatim from FDA Clearance Summary)

The SmartStep System is intended to sense the amount of weight applied to the plantar surface of the foot during rehabilitation and to alert the user and/or therapist with an alarm when the weight exceeds a pre-selected value. The device is indicated for use in adults undergoing rehabilitation who require monitoring of partial weight-bearing on the lower extremity. It is indicated for Prescription Use under the supervision of a physician or physiotherapist.

Device Description (from FDA Clearance Summary)

The SmartStep System consists of three components: a flexible pressure-sensing insole worn inside the patient's shoe, a belt-worn Control Unit that processes sensor data and generates alerts, and PC-based Software for clinical review and threshold configuration. The system is designed to sense the weight applied to the plantar surface of the foot during rehabilitation and to alert the user and/or therapist with an alarm when the weight exceeds a pre-selected value. The device complies with IEC 60601-1, IEC 60601-1-2, and AAMI/ISO 14971-1 (U.S. Food and Drug Administration, "K060150").

Calibration Method (from FDA Clearance Summary)

The SmartStep System uses a user-adjustable weight threshold set via the PC software application prior to use. The clinician or therapist enters the patient's prescribed weight-bearing limit, and the system generates an audible alarm when the measured plantar force exceeds that threshold during rehabilitation exercises. The threshold is referenced against the patient's prescribed partial weight-bearing limit, and both approaches (SmartStep and NeuroFoot) set a single reference threshold per session. The SmartStep's threshold is instantaneous-force-based, while the NeuroFoot's threshold is duration-based, consistent with the sustained-loading etiology of neuropathic ulcers.

Operating Manual Availability

The SmartStep operating manual is not publicly available through the FDA 510(k) Premarket Notification Database, the Andante Medical Devices Ltd. website, or standard commercial channels. The substantial equivalence comparison presented in Table 5 of this report is based on information available in the FDA clearance summary (K060150), which describes the device's intended use, calibration parameters, and component architecture but does not include the full instructions for use. If the operating manual becomes available through a Freedom of Information Act (FOIA) request to FDA or through direct contact with the manufacturer, the predicate comparison should be updated to include a detailed calibration methodology comparison, recalibration interval requirements, sensor maintenance procedures, and end-of-life criteria.

5. Appendix B: Bill of Materials with Vendor Certification Status

The following table lists each component in the NeuroFoot bill of materials, the vendor or manufacturer, the vendor's known quality management certification, and the incoming inspection method applied at Stage 1 of the assembly process. The ISO certification status was determined from publicly available vendor documentation as of April 2026. None of the component vendors hold ISO 13485 medical device quality management certification. All incoming inspection is therefore performed independently by the assembly team per the procedures defined in Section 1.2.d.

Component	Vendor / Manufacturer	Vendor Cert.	Qty	Stage 1 Incoming Inspection
RP-C Thin Film FSR (20g-2kg)	MakerHawk / Shenzhen Ligan Technology Co., Ltd.	ISO 9001	16	Physical inspection for creases, delamination, or sensing element damage. Pre-solder resistance test: near-infinite with no load, measurable drop when pressed.
Arduino Nano 33 BLE Sense Rev2	Arduino S.r.l.	ISO 9001	1	Confirm power-on and BLE discoverability. Verify board revision and pin count.
CD74HC4067 16-Ch Analog Multiplexer	Texas Instruments	ISO 9001	1	Confirm correct part number and pin count. Verify 3.3V logic compatibility.
BMI270 6-Axis IMU (onboard)	Bosch Sensortec	ISO 9001	1 (onboard)	Verified during Stage 3 electrical test. Confirm accelerometer and gyroscope data on all six axes.
HS300x Temp/Humidity Sensor (onboard)	Renesas Electronics	ISO 9001	1 (onboard)	Verified during Stage 3 electrical test. Confirm initialization (up to 3 retries) and valid temp/humidity readings.
10 kΩ Pull-Down Resistors	Various (generic passive)	None documented	16	Verify resistance value with multimeter. Confirm 10 kΩ (not 100 kΩ).
LiPo Battery (3.7V)	Various	None documented	1	Inspect for charge level, swelling, and connector compatibility. Verify polarity.
Custom FR4 PCB	PCB fabrication house (e.g., JLCPCB)	ISO 9001	1	Check for visible trace damage, lifted pads, or contamination. Confirm footprint matches current revision.

EVA Foam Sandal (2-layer)	Various (commercial sandal blank)	None documented	1	Confirm correct size variant. Verify durometer, connector compatibility, and freedom from material defects.
Buzzer / Vibration Motor	Various	None documented	1 each	Verified during Stage 3 electrical test. Confirm alert output activates correctly.
FPC Breakout Board / Connectors	Various	None documented	As needed	Visual inspection for damage. Verify pin compatibility.
Conformal Coating	Various (e.g., MG Chemicals)	None documented	As needed	Verify product within shelf life. Confirm compatibility with FR4 and solder.
Epoxy Adhesive	Various	None documented	As needed	Verify product within shelf life. Confirm structural grade suitable for repeated loading.
Waterproof Foam Resealing Adhesive	Various	None documented	As needed	Verify waterproof rating. Confirm biocompatibility for skin-contact application.
USB-C / Magnetic Charging Connector	Various	None documented	1	Verify connector type and polarity. Test charging function during Stage 9 final QC.

All primary electronic components (FSR sensors, Arduino, multiplexer, onboard IMU and environmental sensor) are manufactured under ISO 9001 general quality management systems. No component vendor holds ISO 13485 medical device quality management certification. Passive components (resistors, battery cells), consumables (adhesives, coatings), and structural materials (foam, connectors) are sourced from vendors with no documented quality certification. This gap is mitigated through mandatory incoming inspection at Stage 1, pre-solder FSR testing, Stage 3 pre-embedding electrical verification, and Stage 9 final quality inspection. If the NeuroFoot system advances toward commercial production, a formal supplier qualification process per ISO 13485 Section 7.4 should be established, including documented supplier audits, an approved supplier list, and purchasing specifications with acceptance criteria independent of vendor certification status.